Installation and operating instructions

MTSE

MTSE
Installation and operating instructions
Other languages
http://net.grundfos.com/qr/i/99688961



GRUNDFOS

MTSE

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Original installation and operating instructions

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1. General information



Read this document before you install the product. Installation and operation must comply with local regulations and accepted codes of good practice.



These installation and operating instructions are a supplement to the installation and operating instructions for the corresponding standard MTS pumps. For instructions not mentioned specifically in this manual, see the installation and operating instructions for the standard pump.

1.1 Related instructions

Title	Model	QR code	Publication number	Link
MTS [1]	C		98189180	*http://net.grundfos.com/qr/i/98189180*
MTS [2]	C		98189187	*http://net.grundfos.com/qr/i/98189187*
MTS [2]	K		93225743	*http://net.grundfos.com/qr/i/93225743*

1) EU version.

2) Asia version.

1.2 Hazard statements

The symbols and hazard statements below may appear in Grundfos installation and operating instructions, safety instructions and service instructions.



DANGER

Indicates a hazardous situation which, if not avoided, will result in death or serious personal injury.



WARNING

Indicates a hazardous situation which, if not avoided, could result in death or serious personal injury.



CAUTION

Indicates a hazardous situation which, if not avoided, could result in minor or moderate personal injury.

The hazard statements are structured in the following way:



SIGNAL WORD

Description of the hazard

Consequence of ignoring the warning

- Action to avoid the hazard.

1.3 Notes

The symbols and notes below may appear in Grundfos installation and operating instructions, safety instructions and service instructions.



Observe these instructions for explosion-proof products.



A blue or gray circle with a white graphical symbol indicates that an action must be taken.



A red or gray circle with a diagonal bar, possibly with a black graphical symbol, indicates that an action must not be taken or must be stopped.



If these instructions are not observed, it may result in malfunction or damage to the equipment.



Tips and advice that make the work easier.

1.4 Abbreviations and definitions

AI	Analog input.
AL	Alarm, out of range at lower limit.
AO	Analog output.
AU	Alarm, out of range at upper limit.
CIM	Communication interface module.
Current sinking	The ability to draw current into the terminal and guide it towards earth in the internal circuitry.
Current sourcing	The ability to push current out of the terminal and into an external load which must return it to earth.

DI	Digital input.
DO	Digital output.
ELCB	Earth leakage circuit breaker.
FM	Functional module.
GDS	Grundfos Digital Sensor, factory-fitted.
GENIbus	Proprietary Grundfos fieldbus standard.
GFCI	Ground fault circuit interrupter.
GND	Protective earth.
Grundfos Eye	Status indicator light.
LIVE	Low voltage with the risk of electric shock if the terminals are touched.
OC	Open collector: Configurable open-collector output.
PE	Protective earth.
RCCB	Residual-current circuit breaker.
RCD	Residual-current device.
SELV	Safety extra-low voltage. A voltage that cannot exceed ELV under normal conditions and under single-fault conditions, including earth faults in other circuits.
STO	Safe Torque Off. A sub safety function, where a drive does not actively generate any torque and coasts freely.

2. Product introduction

2.1 Product description

MTSE pumps are MTS pumps with an E-motor. MTSE pumps belong to the so-called E-pump family and are referred to as E-pumps. MTSE pumps are mounted with frequency-controlled permanent-magnet MGE motors for three-phase power supply connection. The motors incorporate a PI controller.

You can connect the motors to a signal from an external sensor and a setpoint signal enabling control in closed loop. You can also use the motors for an open-loop system in which the setpoint signal is used as a speed control signal.

The motors incorporate an operating panel which is available in various versions.

Detailed motor settings are made with Grundfos GO. Furthermore, you can read important operating parameters via Grundfos GO.

The motors incorporate a functional module. The functional module is available in various versions with different inputs and outputs.

You can fit the motors with a Grundfos add-on communication interface module (CIM). The module enables data transmission between the motor and an external system. The module communicates via fieldbus protocols.

You can connect several motors together via radio or bus communication to create a multimotor system.

2.2 Pumps without a factory-fitted sensor

The pumps have a built-in PI controller and can be set for an external sensor enabling the control of the following parameters:

- constant pressure
- constant differential pressure
- constant temperature
- constant differential temperature
- constant flow rate
- constant level
- constant curve
- constant other value.

The pumps are factory-set to constant-curve control mode. You can change the control mode with the Grundfos GO, HMI 300, HMI 301 or via the Grundfos GO Link.

2.3 Intended use

The product is intended for machines with a constant torque characteristic, such as screw-spindle pumps and other positive displacement pumps.

2.3.1 Intended use in the United Kingdom

For intended use of the product in the United Kingdom, see the appendix.

2.4 Identification

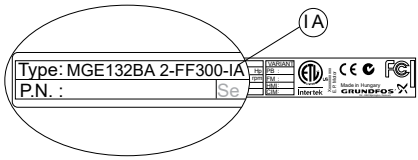
2.4.1 Identification of the pump model

Identify the pump by the nameplate on the pump. See description of the nameplate and type key in the related installation and operating instructions.

2.4.2 Identification of the motor model

Identify the motor by means of the nameplate on the terminal box.

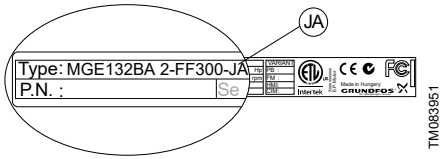
Model I



TM086655

Motor [kW]	3 × 200-240 V	3 × 380-500 V
1.5	•	•
2.2	-	•

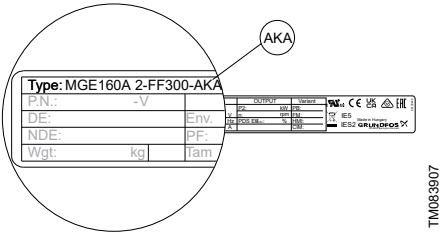
Model J



TM083951

Motor [kW]	3 × 200-240 V	3 × 380-500 V
2.2	•	-
3	•	•
4	•	•
5.5	•	•
7.5	-	•
11	-	•

Model K



TM083907

Motor [kW]	3 × 200-240 V	3 × 380-480 V	3 × 400 - 480 V
7.5	•	-	-
11	•	-	-
15	-	•	-
18.5	-	•	-
22	-	•	-
26	-	-	•

2.4.3 Identification of the functional module

You can identify the fitted module in one of the following ways:

Grundfos GO

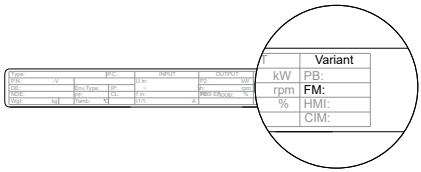
You can identify the functional module in the **Fitted modules** menu under **Status**.

Motor display

For motors fitted with the HMI 300 or 301 operating panel, you can identify the functional module in the **Fitted modules** menu under **Status**.

Motor nameplate

You can identify the fitted module by means of the data on the motor nameplate.



TM082851

Functional module variants:

- FM300
- FM310
- FM311 [3]

3) Without Bluetooth (BLE).

Related information

5.2.8 Functional modules

2.4.4 Identification of the operating panel

You can identify the operating panel in one of the following ways:

Grundfos GO

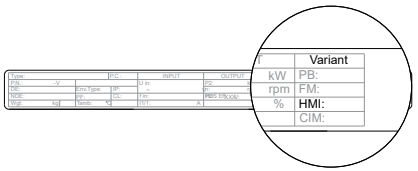
You can identify the operating panel in the **Fitted modules** menu under **Status**.

Motor display

For motors fitted with the HMI 300 or 301 operating panel, you can identify the operating panel in the **Fitted modules** menu under **Status**.

Motor nameplate

You can identify the operating panel by means of the data on the motor nameplate.



TM082852

Operating panel variants

- HMI 200
- HMI 201 [4]

- HMI 300
- HMI 301 [4)]

4) For motors without a radio module.

Related information

7.1 User interfaces

2.5 GLoWpan

CAUTION
Radiation
Minor or moderate personal injury

- Locate the product at a minimum distance of 20 cm from any body parts. Human tissue may be heated by RF energy.

Installers and end users must be provided with these installation and operating instructions and operating conditions for satisfying RF exposure compliance.

The product incorporates a class 1 radio module for remote control. You can use the module anywhere in the EU without restrictions.

For installation in the USA and Canada, see the appendix.

Via the built-in radio module, the product can communicate with other MGE motors.

Grundfos will support the product with security updates for at least 2 years from production of the unit.

GLoWpan information

Frequency of operation	2405-2480 MHz
Modulation type	GP O-QPSK
Data rate	1 Mbps
Transmit power	5 dBm EIRP with internal antenna

2.6 Bluetooth

The product incorporates a built-in Bluetooth (BLE) module for remote control via Grundfos GO. You can use the module anywhere in the EU without restrictions.

For installation in the USA and Canada, see the appendix.

Grundfos will support the product with security updates for at least 2 years from production of the unit.

Bluetooth information

Frequency of operation	2400 - 2483.5 MHz
Modulation type	GFSK
Data rate	2 Mbps
Transmit power	5 dBm EIRP with internal antenna

2.7 Battery

A Li-ion battery is fitted in the FM310 and FM311 functional modules.

The Li-ion battery complies with the Battery Directive (2006/66/EC). The battery does not contain mercury, lead or cadmium.

WARNING
Intoxication or risk of chemical burn
Death or serious personal injury

- The battery can cause severe or fatal injuries in 2 hours or less if it is swallowed or placed inside any part of the body. In such an event, seek medical attention immediately.

- The replacement or servicing of batteries must be carried out by a qualified person.
- The battery contained within this product, whether new or used, is hazardous and is to be kept away from children.

2.8 Safe Torque Off (STO) function

The Safe Torque Off (STO) is a safety function to stop the motor from turning without actively braking it. It follows the definition by EN61800-5-2.

For instructions on how to activate and operate the Safe Torque Off (STO) function, read the following installation and operating instructions.

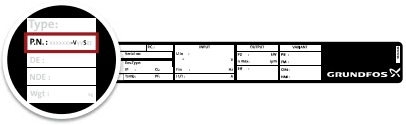
Safe Torque Off
Installation and operating instructions
http://net.grundfos.com/qr/i/92916582

2.8.1 Identification of the Safe Torque Off (STO) function

The version of the Safe Torque Off (STO) function is marked on the nameplate, after the product version number.

The Safe Torque Off (STO) functionality is only available for MGE, MLE motors having an STO version number.

The Safe Torque Off (STO) version number is shown below as **Szz**, where **zz** marks the version. For product without STO the **zz** segment will be blank.



TM084339

The Safe Torque Off (STO) safety function cannot be retrofitted to older motors.

2.9 Cybersecurity

The product must only be connected to protected network subnets with strict access control.

According to the standard EN 18031-1:2024, the RJ45 connector used for connecting to IP networks is considered an exposed network interface, along with the GENIbus TCP protocol used on it, which is considered an exposed service.

2.10 Network interfaces and services

In the factory default state, the product exposes the following network interfaces:

Interface	Description
RJ45	Wired Ethernet / IP connectivity

The following services are exposed by the product over its network interfaces in the factory default state.

Interface	Service	Description
RJ45	GENIpro TCP	It is used for connecting the product to a service tool on an engineering workstation. Pairing is required before communication can take place. The GENIpro TCP pairing state is persistent.

3. Receiving the product

3.1 Transporting the product

WARNING
Falling objects
Death or serious personal injury
- Secure the product during transport to prevent it from tilting or falling down.

WARNING
Crushing of feet
Minor or moderate personal injury
- Wear safety shoes when moving the product.

When you transport the product, pay attention to the following:
- Motors from 2.2 to 5.5 kW: Do not stack more than two motors in the original packaging.
- Motors from 5.5 to 11 kW: Do not stack the motors.

3.2 Inspecting the product

Before installing the product, do the following:

1. Check that the product is as ordered.
   If the product is not as ordered, contact the supplier.
2. Check that no visible parts have been damaged.
   If any visible parts have been damaged, contact the transport company.

3.3 Lifting the product

WARNING
Falling objects
Death or serious personal injury
- Do not use the motor eyebolts to lift the entire pump if the pump is fitted with a motor of another make than Grundfos MG and MGE.
- Follow the lifting instructions.
- Use lifting equipment that is approved for the weight of the product.
- Persons must keep a safe distance from the product during lifting operations.
- Wear personal protective equipment.

For lifting instructions, see the related installation and operating instructions for the pump.

Related information

5.1 Mounting the product

9. Servicing the product

4. Installation requirements

4.1 Installing the product outdoors or in areas with high humidity

WARNING
Fire hazard
Death or serious personal injury

- In high humidity environments where condensation can occur, connect the product permanently to the mains supply and activate the standstill heating function.

To maintain the cURus mark, additional requirements apply to the equipment. See the appendix concerning installation in the USA and Canada.

Do not expose the product to UV radiation.

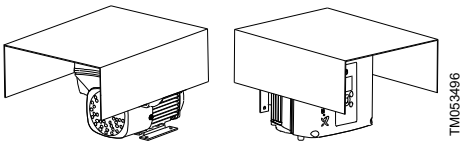
To avoid condensation, the drive must be continuously energized due to the application of heat, with interruptions such that cooling to the point of condensation does not occur.

If you install the product outdoors or in areas with a high humidity, take the following action to avoid condensation on the electronic components.

- Provide the product with a suitable cover.

  The cover must be large enough to ensure that the product is not exposed to direct sunlight, UV radiation, rain or snow. Grundfos does not supply covers.

  When fitting a cover to the product, observe the instructions for adequate cooling.



- Open the drain holes in the product.

  When you open the drain hole, the enclosure class of the motor will be lower than standard.

- Connect the product permanently to the mains supply. In areas with a high humidity, activate the built-in standstill heating function.

  If you install the motor in moist surroundings or areas with a high humidity, ensure that the bottom drain hole is open. As a result, the motor becomes self-venting, allowing water and humid air to escape. When you open the drain hole, the enclosure class of the motor will be lower than standard.

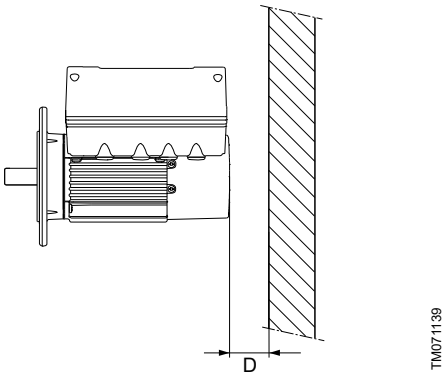
Related information

5.1.2 Drain holes

4.2 Minimum space

4.2.1 Cooling the motor

- Install the motor allowing a distance of minimum 50 mm (D) between the end of the fan cover and the wall or another fixed object.



- Position the product with sufficient space around.
- Make sure that the temperature of the cooling air does not exceed 50 °C.
- Keep cooling fins and fan blades clean.

Related information

5.1 Mounting the product

5. Mechanical installation

5.1 Mounting the product

WARNING
Crushing of feet
Death or serious personal injury

- Fasten the pump securely to a solid and even foundation according to the specifications in the installation and operating instructions for the pump.
- Follow the lifting instructions.

CAUTION
Radiation
Minor or moderate personal injury

- Locate the product at a minimum distance of 20 cm from any body parts. Human tissue may be heated by RF energy.

CAUTION
Tilting object
Death or serious personal injury

- Keep your distance when lifting the motor, as the motor may tilt slightly to the sides.

Installation-related work on the product must only be performed by qualified persons.

The product must be installed in a location with access control to prevent unauthorized access to the product.

For lifting instructions, see the related installation and operating instructions for the pump.

Related information

3.3 Lifting the product
4.2.1 Cooling the motor

5.1.1 Handling the product

Observe local regulations concerning limits for manual lifting or handling. The weight of the product is stated on the nameplate.

CAUTION
Back injury
Minor or moderate personal injury

- Use lifting equipment.

CAUTION
Crushing of feet
Minor or moderate personal injury

- Wear safety shoes.
- Attach lifting equipment to the motor eyebolts.

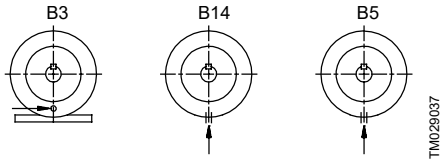
Do not lift the product by the terminal box.

5.1.2 Drain holes

The motor has a plugged drain hole on the drive side. The drain hole is placed in the flange on the drive side. You can turn the flange 90° to both sides or 180°.

With the drain hole open, the motor becomes self-venting, allowing water and humid air to escape.

When you open the drain hole, the enclosure class of the motor is lower than standard.



Related information

4.1 Installing the product outdoors or in areas with high humidity

5.1.3 Changing the position of the operating panel

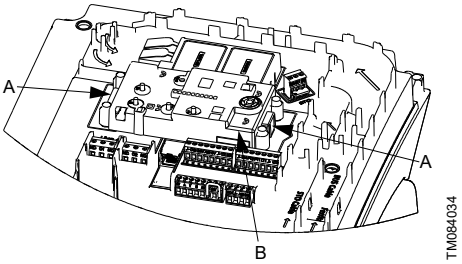
WARNING
Electric shock
Death or serious personal injury

- Switch off the power supply to the product including the power supply for the signal relays. Wait at least 5 minutes before you make any connections in the terminal box.

You can turn the operating panel 180°. Follow the instructions.

1. Loosen the four screws on the terminal box cover.
2. Remove the terminal box cover.

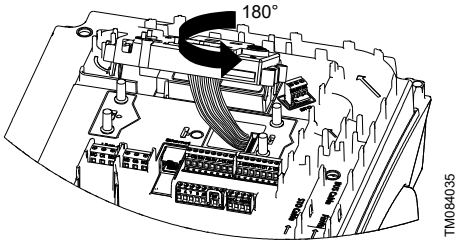
3. Press and hold in the two locking tabs (A) while gently lifting the plastic cover (B).



Lifting the plastic cover, shown on a Model J motor

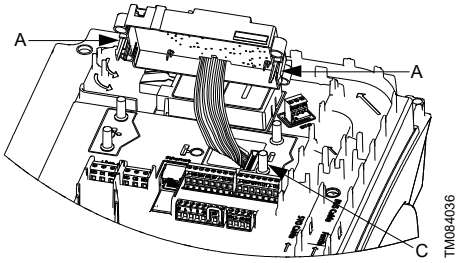
4. Turn the plastic cover 180°.

Do not twist the cable more than 90°.



Turning the plastic cover, shown on a Model J motor

5. Position the plastic cover correctly over the four rubber pins (C). Make sure that the locking tabs (A) are placed correctly.



Positioning the plastic cover, shown on a Model J motor

6. Fit the terminal box cover and cross-tighten the four screws to 5 Nm.

Make sure that the terminal box cover is aligned with the orientation of the operating panel.

5.2 Electrical connection

WARNING
Electric shock
Death or serious personal injury

- Switch off the power supply to the product including the power supply for the signal relays. Wait at least 5 minutes before you make any connections in the terminal box. Make sure that the power supply cannot be switched on accidentally.
- Check that the supply voltage and frequency correspond to the values stated on the nameplate.
- Connect the pump to an external power switch close to the pump and to a motor-protective circuit breaker. Make sure you can lock the power switch in OFF position (isolated). Type and requirements as specified in EN 60204-1, 5.3.2.

If the power cable is damaged, it must be replaced by the manufacturer, the manufacturer's service partner or a similarly qualified person.

The user or the installer is responsible for correct earthing and protection according to local regulations.

All electrical connections must be carried out by qualified persons.

Make sure to fill the pump with water before the power is switched on. Follow the instructions for the pump.

5.2.1 Connecting an external switch

We recommend that you connect the product to an external switch.

1. Connect the switch via terminals 2 (DI1) and 6 (GND).
   A jumper is added from factory.
2. Enable the **External stop** function.
   Default setting from factory.

5.2.2 Electrical supply systems

Power supply network and earthing systems



If you want to supply the product through an IT network, make sure that you have a suitable product variant. If you are in doubt, contact Grundfos.

The internal EMC filter remains connected, and subsequently no reduced leakage current variant is available.

Supply line types

The product is not suitable for use on corner earthed grids in installations more than 2000 m above sea level.

- TN-S earthing system
- TN-C earthing system
- TN-C-S earthing system
- TT earthing system

5.2.3 Protection against electric shock, indirect contact



WARNING
Electric shock
Death or serious personal injury

- Connect the product to protective earth and provide protection against indirect contact in accordance with local regulations.

Protective-earth conductors must have a yellow and green (PE) or yellow, green and blue (PEN) colour marking.

5.2.4 Protection against mains voltage transients

The product is protected against mains voltage transients in accordance with EN 61800-3.

5.2.5 Motor protection

The product incorporates thermal protection against slow overloading and blocking. No external motor protection is required.

The product includes load and speed sensitive motor overload protection.

5.2.6 Cable requirements



WARNING
Electric shock
Death or serious personal injury

- Switch off the power supply to the product including the power supply for the signal relays. Wait at least 5 minutes before you make any connections in the terminal box.
- Follow the wiring diagrams and local regulations.
- Use branch-circuit protection fuses.
- Comply with local regulations as to cable cross-sections.
- Use the recommended fuse size.
- Connect the cables to terminals by applying the recommended tightening torque.



WARNING
Fire hazard
Death or serious personal injury

- Comply with local regulations as to cable cross-sections.
- Use the recommended fuse size.
- Connect the cables to terminals by applying the recommended tightening torque.



Make sure that the cables are secured with cable glands providing strain relief.

Cable cross-section
Model I

Supply voltage [V]	Power P2 [kW]	Nominal current [A]	Cable cross-section [mm^2]	Cable cross-section [AWG]
3 × 200-240	1.5	5.4 - 4.6	1.5 - 10	16-8
3 × 380-500	1.5	2.9 - 2.4	1.5 - 10	16-8
	2.2	4.1 - 3.4	1.5 - 10	16-8

Model J

Supply voltage [V]	Power P2 [kW]	Nominal current [A]	Cable cross-section [mm²]	Cable cross-section [AWG]
3 × 200-240	2.2	7.8 - 6.5	2.5	14
	3.0	10.5 - 8.8	2.5	14
	4.0	14.1 - 11.8	4	12
	5.5	19.6 - 16.3	6	10
3 × 380-500	3.0	5.8 - 4.6	1.5	14
	4.0	7.7 - 6.0	2.5	14
	5.5	10.5 - 8.4	2.5	14
	7.5	14.1 - 11.1	4	12
	11.0	20.3 - 16.0	6	10

Model K

Supply voltage [V]	Power P2 [kW]	Nominal current [A]	Cable cross-section [mm²]	Cable cross-section [AWG]
3 × 200-240	7.5	25.6 - 21.4	10	8
	11	37.4 - 31.4	10	8
3 × 380-480	15	26.7 - 22	6	8
	18.5	33 - 27.8	10	8
	22	39.2 - 31.5	10	8
3 × 400-480	26	43.8 - 37.6	16	6

Conductor types

Stranded or solid copper conductors.

Conductor temperature ratings

Temperature rating for conductor insulation: 60 °C (140 °F).

Temperature rating for outer cable sheath: 75 °C (167 °F).

Related information

12.5.7 Torques for terminals

5.2.6.1 Cable entries

The cable entries are fitted with blanking plugs from the factory. See the cable entry sizes in the section on other technical data.

5.2.6.2 Cable glands

See the list of cable gland sizes in relation to motor sizes in the section on other technical data.

It is recommended to use a cable gland M20 or M40 as applicable with IP 66 rating and suitable for cable strain relief.

After installation, all M20 openings must be closed by means of the delivered blind plugs to maintain the IP 55/66 rating.

Related information

12.5.1 Ecodesign Directive

12.5.6 Cable gland sizes

5.2.6.3 Three-phase connections

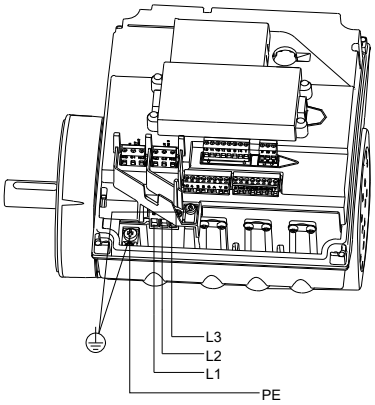
The cables in the terminal box must be as short as possible. However, the separated protective-earth conductor must be so long that it is the last one to be disconnected in case the cable is inadvertently pulled out of the cable entry.

To maintain the cURus mark, additional requirements apply to the equipment. See the appendix concerning installation in the USA and Canada.

To avoid loose connections, ensure that the terminal block for L1, L2 and L3 is pressed home in its socket when the power cable has been connected.

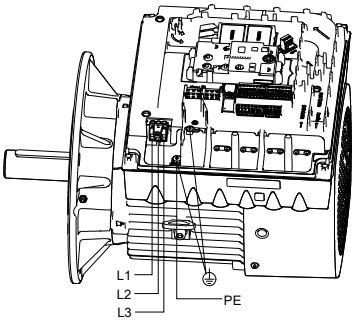
Check that the supply voltage and frequency correspond to the values stated on the nameplate.

Power supply connection on a three-phase product



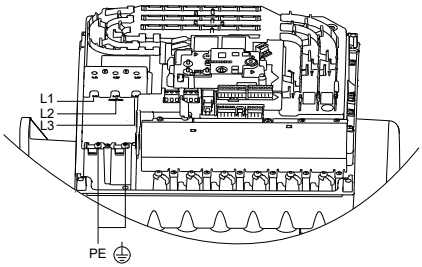
TM053495

Model I



TM084037

Model J



TM082860

Model K

Pos.	Description
L1	Phase 1
L2	Phase 2
L3	Phase 3
PE	Protective earth

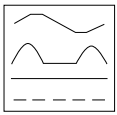
5.2.7 Additional protection

5.2.7.1 Residual-current circuit breakers

WARNING
Electric shock
Death or serious personal injury

- This product can cause a DC current in the protective-earth conductor. If a residual current-operated protective (RCD) or monitoring (RCM) device is used for protection in case of direct or indirect contact, only an RCD or RCM of Type B is allowed on the supply side of this product.

The residual-current circuit breaker must be marked.



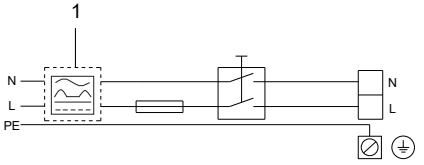
TM066230

Take into account the total leakage current of all the electrical equipment in the installation.

This product may cause a direct current in the protective-earth conductor.

Connection example for single-phase supply

The figure shows an example of a mains-connected single-phase motor with a main switch, a backup fuse and a residual-current circuit breaker, type B.

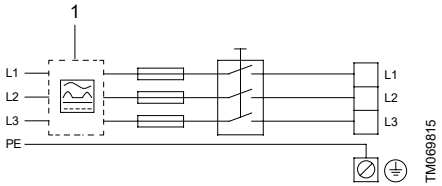


TM069814

Pos.	Description
1	Residual-current circuit breaker, type B
N	Neutral
L	Phase
PE	Protective earth

Connection example for three-phase supply

The figure shows an example of a mains-connected three-phase motor with a main switch, a backup fuse and a residual-current circuit breaker, type B.



Pos.	Description
1	Residual-current circuit breaker, type B
L1	Phase 1
L2	Phase 2
L3	Phase 3
PE	Protective earth

5.2.7.2 Overvoltage and undervoltage protection

Overvoltage and undervoltage may occur in case of unstable power supply or a faulty installation. The product stops if the voltage falls outside the permissible voltage range. The product restarts automatically when the voltage is within the permissible voltage range. The product requires no additional protection relay.

The product is protected against transients from the power supply according to EN 61800-3. In areas with high lightning intensity, we recommend external lightning protection.

Overvoltage category:

The product is approved for Overvoltage category III rating.

5.2.7.3 Overload protection

The motor-current protection settings are fixed for each motor variants. The settings ensure that the motor is protected against overtemperature in all operating states with regard to supply voltage and shaft load, including a blocked shaft.

The motors are current controlled and will respond by reducing the speed if the shaft load increases more than 10 % of the nominal load.

If the shaft load forces the speed down to minimum speed, the motor shuts down.

A sudden increase in the motor current caused by a fault where the peak of the motor current is increased 60 % above nominal will cause the motor to shut down within 0.5 ms.

The product requires no additional protection.

5.2.7.4 Overtemperature protection

The motor is thermally protected by a temperature measurement in the drive. It can handle the lack of airflow over the motor in case the fan cover is blocked. It also means that the protection has a built-in memory retention.

The time from start to shutdown due to overtemperature is therefore always longer when starting at a motor temperature close to the ambient temperature compared with restarting after a shutdown due to overtemperature.

5.2.7.5 Protection against phase unbalance

Phase unbalance on the power supply must be minimised. The three-phase motor must be connected to a power supply with a quality corresponding to IEC 60146-1-1, class C. This also ensures long life of the components.

5.2.7.6 Short-circuit current

The product's electronic power output short-circuit protection circuitry meets the requirements of IEC 60364-4-41:2005/AMD1:-, Clause 411.

If a short circuit occurs, the pump can be used on a power supply delivering not more than 5000 RMS symmetrical amperes, 600 V maximum.

5.2.8 Functional modules

The functional modules are different types of add-on boards containing various types of input and output terminals for the user to connect different types of sensors, for example switches and relays.

The product can only contain one functional module at the time.

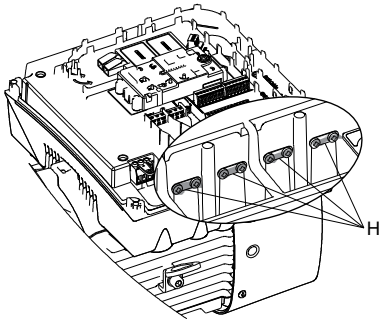
The following functional modules are available:

- FM300
- FM310
- FM311 [5)]

5) Without Bluetooth (BLE).

The selection of module depends on the application and the required number of inputs and outputs.

The screen of signal cables and bus connection cables must be connected to ground via one of the earth clamps (H).



Related information

2.4.3 Identification of the functional module

5.2.10 Signal cables

5.2.8.1 Model I

5.2.8.2 Functional module, FM300

Inputs and outputs

The module has these connections:

- three analog inputs
- one analog output
- two dedicated digital inputs
- two configurable digital inputs or open-collector outputs
- Grundfos Digital Sensor input and output
- LiqTec sensor inputs
- two signal relay outputs
- GENIbus connection.

The inputs and outputs are internally separated from the power supply-conducting parts by reinforced insulation and galvanically separated from other circuits. All control terminals are supplied with protective extra-low voltage (PELV), ensuring protection against electric shock.

Signal relay 1

LIVE: You can connect supply voltages up to 250 VAC to the output.

PELV: The output is galvanically separated from other circuits. Therefore, you can connect the supply voltage or protective extra-low voltage to the output as desired.

Signal relay 2

PELV: The output is galvanically separated from other circuits. Therefore, you can connect the supply voltage or protective extra-low voltage to the output as desired.

Connection terminals for the power supply

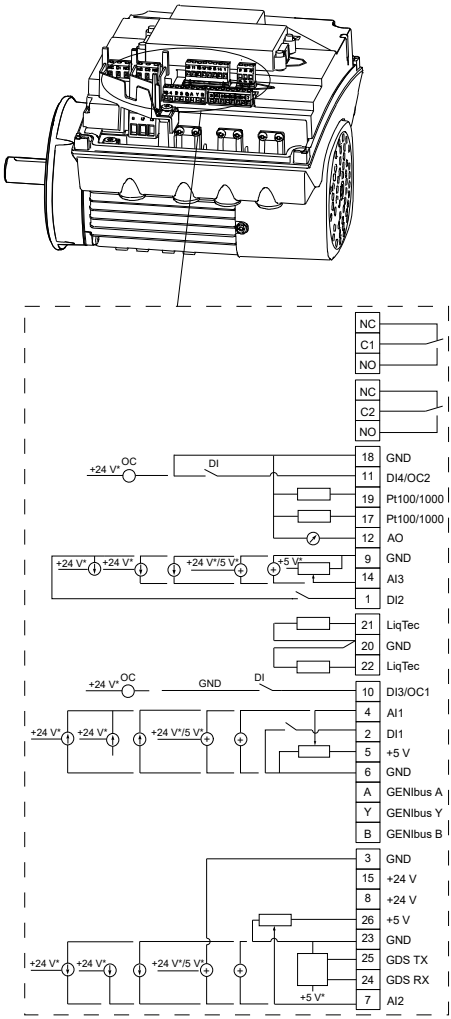
Phases	Terminals
Single-phase	N, PE, L
Three-phase	L1, L2, L3, PE

Connection terminals for inputs and outputs



DANGER
Electric shock
Death or serious personal injury

- Make sure that the wires to be connected to the connection groups below are separated from each other by reinforced insulation in their entire lengths.



Terminal	Type	Function
NC	Normally closed contact	Signal relay 1. LIVE or PELV.
C1	Common	
NO	Normally open contact	
NC	Normally closed contact	Signal relay 2. PELV only.
C2	Common	
NO	Normally open contact	
18	GND	Protective earth.

Terminal	Type	Function
11	DI4/OC2	Digital input/output, configurable. Open collector: Maximum 24 V resistive or inductive.
19	Not in use	-
17	Not in use	-
12	AO	Analog output: 0-20 mA or 4-20 mA 0-10 V.
9	GND	Protective earth.
14	AI3	Analog input: 0-20 mA or 4-20 mA. 0.5 - 3.5 V, 0-5 V or 0-10 V.
1	DI2	Digital input, configurable.
21	LiqTec sensor input 1	LiqTec sensor input. White conductor.
20	GND	Protective earth. Brown and black conductors.
22	LiqTec sensor input 2	LiqTec sensor input. Blue conductor.
10	DI3/OC1	Digital input/output, configurable. Open collector: Maximum 24 V resistive or inductive.
4	AI1	Analog input: 0-20 mA or 4-20 mA. 0.5 - 3.5 V, 0-5 V or 0-10 V.
2	DI1	Digital input, configurable.  Digital input 1 is factory-set to be start or stop input where an open circuit results in stop. A jumper has been factory-fitted between terminals 2 and 6. Remove the jumper if digital input 1 is to be used as external start or stop or any other external function.
5	+5 V	Power supply to a potentiometer or sensor.
6	GND	Protective earth.
A	GENIbus, A	GENIbus, A (+).
Y	GENIbus, Y	GENIbus, Y (GND).
B	GENIbus, B	GENIbus, B (-).
3	GND	Protective earth.
15	+24 V	Power supply.
8	+24 V	Power supply.
26	+5 V	Power supply to a potentiometer or sensor.
23	GND	Protective earth.

Terminal	Type	Function
25	GDS TX	Grundfos Digital Sensor output.
24	GDS RX	Grundfos Digital Sensor input.
7	AI2	Analog input: 0-20 mA or 4-20 mA. 0.5 - 3.5 V, 0-5 V or 0-10 V.

5.2.8.3 Model J and model K

5.2.8.4 Functional module, FM310 and FM311

Inputs and outputs



The FM311 functional module does not include Bluetooth connection.

The module has these connections:

- three analog inputs
- one analog output
- two dedicated digital inputs
- two configurable digital inputs or open-collector outputs
- Grundfos Digital Sensor input and output
- two Pt100/1000 inputs
- two LiqTec sensor inputs
- two signal relay outputs
- GENIbus/Modbus connection
- two Safe Torque Off (STO) inputs
- Ethernet connection
- Bluetooth (BLE) connection. [6)]

6) FM311 is without Bluetooth.

Signal relay 1

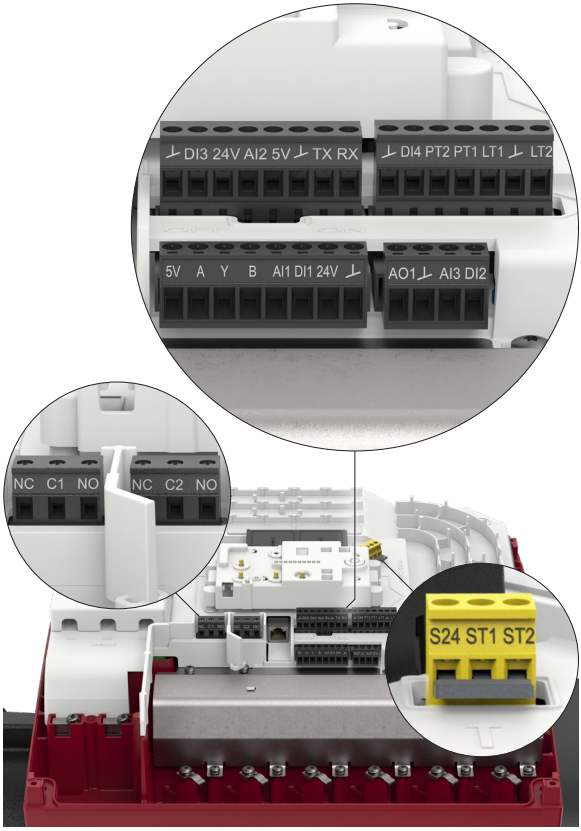
LIVE: You can connect supply voltages up to 250 VAC to the output.

SELV: The output is galvanically separated from other circuits. Therefore, you can connect the supply voltage or safety extra-low voltage to the output as desired.

Signal relay 2

SELV: The output is galvanically separated from other circuits. Therefore, you can connect the supply voltage or safety extra-low voltage to the output as desired.

Terminal overview



TM088957

Terminal	Type	Function
NC	Normally closed contact	Signal relay 1: LIVE or SELV
C1	Common	
NO	Normally open contact	
NC	Normally closed contact	Signal relay 2: SELV only
C2	Common	
NO	Normally open contact	
RJ45	Ethernet	Ethernet communication
GND	GND	Signal ground
DI3	DI3/OC1	Digital input/output, configurable Open collector: Maximum 24 V resistive or inductive
24V	+24 V	Power supply

Terminal	Type	Function
AI2	AI2	Analog input: • 0-20 mA or 4-20 mA • 0.5 - 3.5 V, 0-5 V or 0-10 V.
5V	+5 V	Power supply to a potentiometer or sensor
GND	GND	Signal ground
TX	GDS TX	Grundfos Digital Sensor output
RX	GDS RX	Grundfos Digital Sensor input
GND	GND	Signal ground
DI4	DI4/OC2	Digital input/output, configurable Open collector: Maximum 24 V resistive or inductive
PT2	Pt100/1000 input 2	Pt100/1000 sensor input 2
PT1	Pt100/1000 input 1	Pt100/1000 sensor input 1
LT1	LiqTec sensor input 1	LiqTec sensor input 1 White conductor
GND	GND	Signal ground Brown and black conductors
LT2	LiqTec sensor input 2	LiqTec sensor input 2 Blue conductor
5V	+5 V	Power supply to a potentiometer or sensor
A	GENIbus, A	GENIbus, A (+) / Modbus, D1 (+)
Y	GENIbus, Y	GENIbus, GND / Modbus, GND
B	GENIbus, B	GENIbus, B (-) / Modbus, D0 (-)
AI1	AI1	Analog input: • 0-20 mA or 4-20 mA • 0.5 - 3.5 V, 0-5 V or 0-10 V.
DI1	DI1	Digital input, configurable  Digital input 1 is factory-set to be start or stop input where an open circuit results in stop. A jumper has been factory-fitted between terminals DI1 and GND. Remove the jumper if digital input 1 is to be used as external start or stop or any other external function.
24V	+24 V	Power supply
GND	GND	Signal ground
AO1	AO	Analog output: • 0-20 mA or 4-20 mA • 0-10 V.
GND	GND	Signal ground
AI3	AI3	Analog input: • 0-20 mA or 4-20 mA • 0.5 - 3.5 V, 0-5 V or 0-10 V.
DI2	DI2	Digital input, configurable

Terminal	Type	Function
S24	+24 V (STO)	Power supply to the Safe Torque Off inputs
ST1	STO1	Safe Torque Off - Input 1
ST2	STO2	Safe Torque Off - Input 2

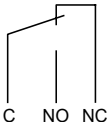
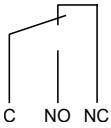
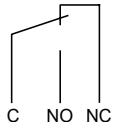
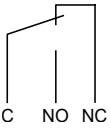
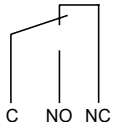
5.2.9 Signal relays

The motor has two outputs for potential-free signals via two internal relays. You can set the signal outputs to **Operation**, **Pump running**, **Ready**, **Alarm** and **Warning**.

The functions of the two signal relays are shown in the table below.

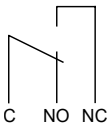
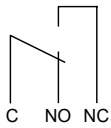
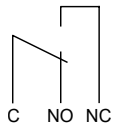
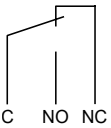
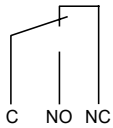
Grundfos Eye is off

The power is off.

Operation	Pump running	Ready	Alarm	Warning	Operating mode
					-

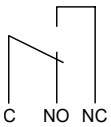
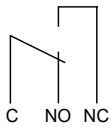
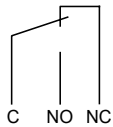
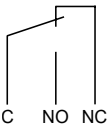
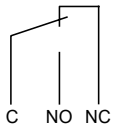
Grundfos Eye is rotating green

The pump runs in **Normal** mode in open or closed loop.

Operation	Pump running	Ready	Alarm	Warning	Operating mode
					**Normal** **Min.** or **Max.**

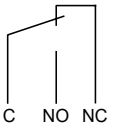
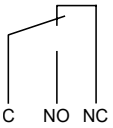
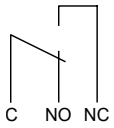
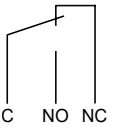
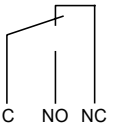
Grundfos Eye is rotating green

The pump runs in **Manual** mode.

Operation	Pump running	Ready	Alarm	Warning	Operating mode
					**Manual**

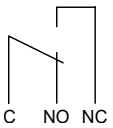
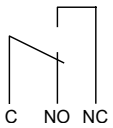
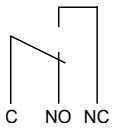
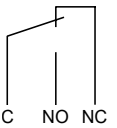
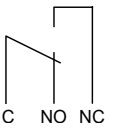
Grundfos Eye is permanently green

The pump is ready for operation but is not running.

Operation	Pump running	Ready	Alarm	Warning	Operating mode
					**Stop**

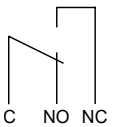
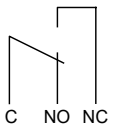
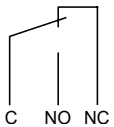
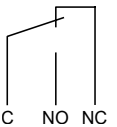
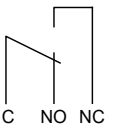
Grundfos Eye is rotating yellow

There is a warning, but the pump is running.

Operation	Pump running	Ready	Alarm	Warning	Operating mode
					**Normal** **Min.** or **Max.**

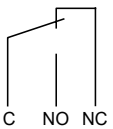
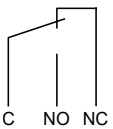
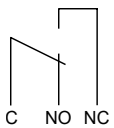
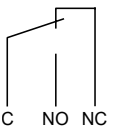
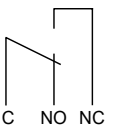
Grundfos Eye is rotating yellow

There is a warning, but the pump is running.

Operation	Pump running	Ready	Alarm	Warning	Operating mode
					**Manual**

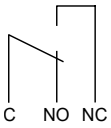
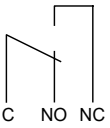
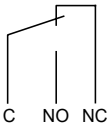
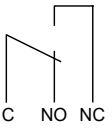
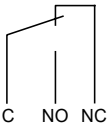
Grundfos Eye is permanently yellow

There is a warning, and the pump is stopped via a **Stop** command.

Operation	Pump running	Ready	Alarm	Warning	Operating mode
					**Stop**

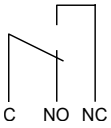
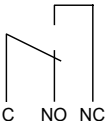
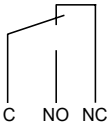
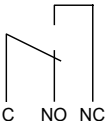
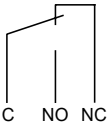
Grundfos Eye is rotating red

There is an alarm, but the pump is running.

Operation	Pump running	Ready	Alarm	Warning	Operating mode
					**Normal** **Min.** or **Max.**

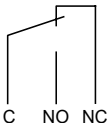
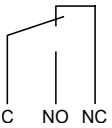
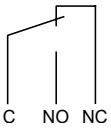
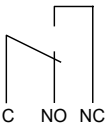
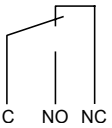
Grundfos Eye is rotating red

There is an alarm, but the pump is running.

Operation	Pump running	Ready	Alarm	Warning	Operating mode
					**Manual**

Grundfos Eye is flashing red

The pump is stopped due to an alarm.

Operation	Pump running	Ready	Alarm	Warning	Operating mode
					**Stop**

Related information

11. Fault finding

5.2.10 Signal cables

Use screened cables with a cross-sectional area of minimum 0.5 mm^2 and maximum 1.5 mm^2 for the external on/off switch, digital inputs, setpoint and sensor signals.

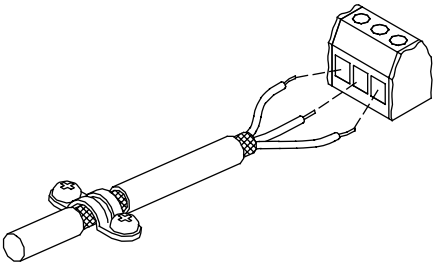
The wires in the motor terminal box must be as short as possible.

Related information

5.2.8 Functional modules

5.2.10.1 Connecting signal cables

1. Connect the screens of the cables to the frame at both ends with good connection. The screens must be as close as possible to the terminals.



2. Connect the signal cables to the terminals.
3. Depending on the model, tighten one or two earth clamp screws.

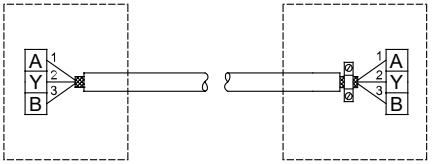
   See the section on functional modules.

5.2.11 Bus connection cable

5.2.11.1 Connecting a 3-core bus cable, GENIbus

For the bus connection, use a screened 3-core cable with a cross-sectional area of minimum 0.5 mm^2 and maximum 1.5 mm^2.

- If the motor is connected to a unit with a cable clamp identical to the one on the product, connect the screen to the cable clamp.
- If the unit has no cable clamp, leave the screen unconnected at this end.



5.2.11.2 Connecting a 3-core bus cable, Modbus

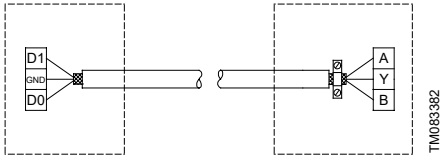
A screened, twisted-pair cable must be used. The cable screen must be connected to protective earth at both ends.

Recommended connection

Terminal	Modbus	Colour code	Data signal
A	D1	Yellow	Positive
B	D0	Brown	Negative
Y	Common/ GND	Grey	Common/ GND

Fitting the cable

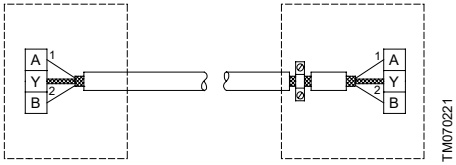
1. Connect the yellow conductor to terminals D1 and A.
2. Connect the brown conductor to terminals D0 and B.
3. Connect the grey conductor to terminals Common/GND and Y.
4. Connect the cable screens to protective earth via the earth clamp.



It is important to connect the screen to protective earth through the earth clamp and to connect the screen to protective earth in all units connected to the bus line.

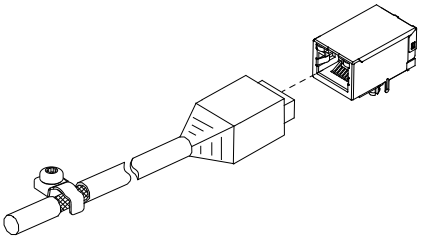
5.2.11.3 Connecting a 2-core bus cable

- Connect a screened 2-core bus cable as follows:



5.2.11.4 Connecting an Ethernet cable

For the USA and Canada, the connection of Ethernet cables must be done by connecting the Ethernet cable screen to an earth clamp on the terminal box, as shown below.



Connecting an Ethernet cable

The Ethernet cable must be rated at least Cat5e/Cat6 with screening.

The recommended Ethernet cable types for earth clamp applications are SF/UTP, S/FTP or SF/FTP, where the cable screen consists of both a braided and a foil screen.

5.2.11.5 Bus signal

The product enables serial communication via an RS-485 input. The communication is carried out according to the Grundfos GENIbus protocol and enables connection to a building management system or another external control system.

Via a bus signal, you can remote-set operating parameters, such as setpoint and operating mode. At the same time, the product can provide status information about important parameters, such as the actual value of the control parameter, input power and fault indications, via the bus.

Contact Grundfos for further information.

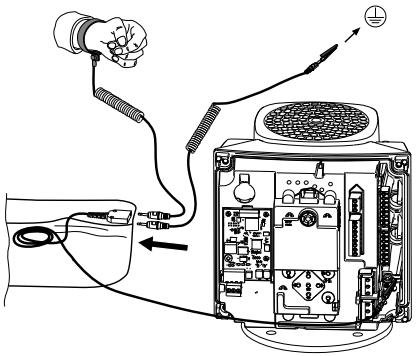
If you use a bus signal, the local settings made via Grundfos GO or the HMI 300 or 301 operating panel will be overruled. In case the bus signal fails, the product will run with the local settings made via Grundfos GO or the HMI 300 or 301 operating panel.

5.2.12 Installing a communication interface module

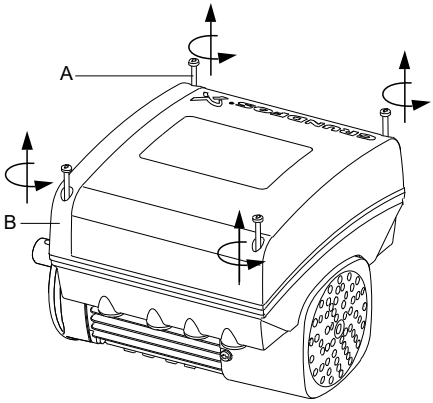
DANGER
Electric shock
Death or serious personal injury

- Switch off the power supply to the product including the power supply for the signal relays. Wait at least 5 minutes before you make any connections in the terminal box. Make sure that the power supply cannot be switched on accidentally.

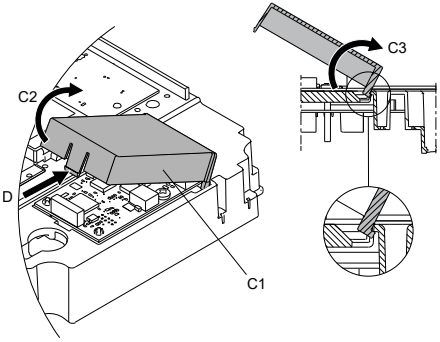
Use an antistatic service kit when handling electronic components. This prevents static electricity from damaging the components.



1. Loosen the four screws (A) and remove the terminal box cover (B).

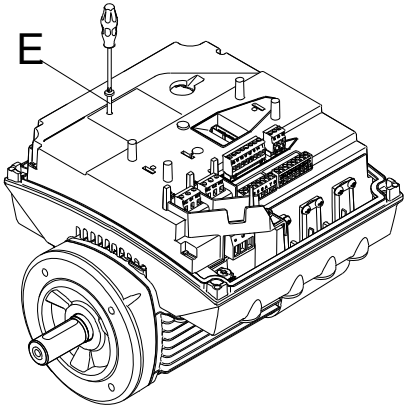


2. Remove the CIM (Communication Interface Module) cover (C1) by pressing the locking tab (D) and lifting the end of the cover (C2). Then lift the cover off the hooks (C3).



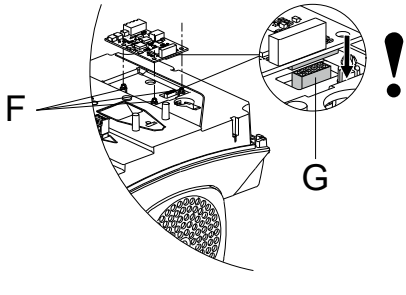
TM069905

3. Remove the screw (E).



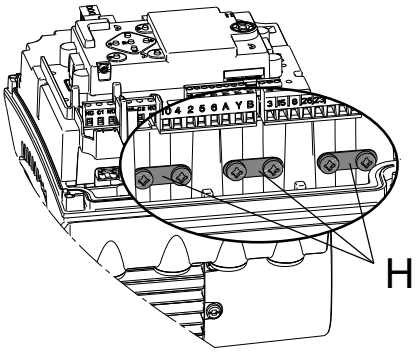
TM069906

4. Fit the module by aligning it with the three plastic holders (F) and the connection plug (G). Press the module home, using your fingers.



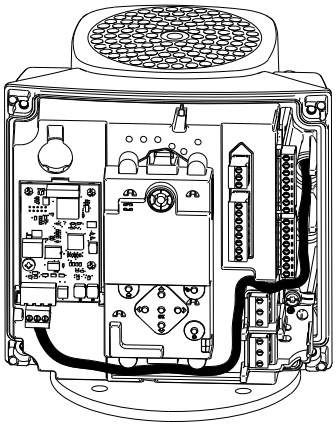
TM069907

5. Fit and tighten the screw (E) to 1.3 Nm.
6. Make the electrical connections to the module as described in the instructions supplied with the module.
7. Connect the cable screens of the bus cables to protective earth via one of the earth clamps (H).



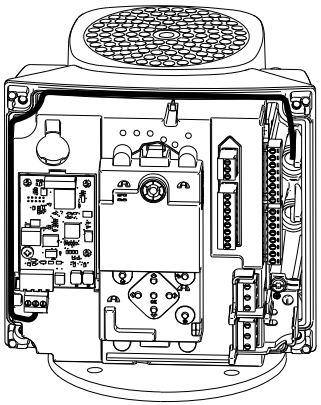
TM069908

8. Route the wires for the module through one of the cable glands.



TM064085

MGE 71, 80, 90

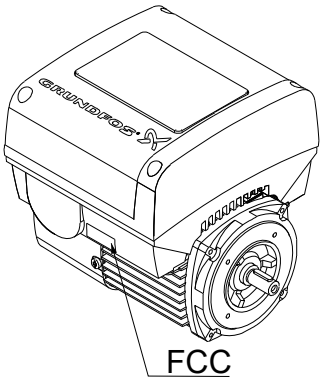


TM070190

MGE 100, 112, 132, 160

9. Fit the CIM cover.

10. If the module is supplied with an FCC label, fix the label on the terminal box.



TM069909

11. Fit the terminal cover and cross-tighten the four mounting screws to 6 Nm.

Make sure that the terminal box cover is aligned with the orientation of the operating panel.

6. Starting up the product

WARNING
Rotating parts
Death or serious personal injury
- Make sure to install the coupling guards before powering on the product.

WARNING
Corrosive liquids
Death or serious personal injury
- Wear personal protective equipment.

WARNING
Toxic liquids
Death or serious personal injury
- Wear personal protective equipment.

CAUTION
Cold surface
Minor or moderate personal injury
- Make sure that no one can accidentally come into contact with cold surfaces. Wear protective gloves.



CAUTION
Hot surface
Minor or moderate personal injury
- Do not touch the product while it is running.



Follow the startup instructions for the pump. See the related installation and operating instructions for the pump.

7. Control functions

7.1 User interfaces



WARNING
Hot surface
Death or serious personal injury
- Touch only the buttons on the operating panel. The product may be very hot.



WARNING
Electric shock
Death or serious personal injury
- If the operating panel is cracked or perforated, replace it immediately. Contact the nearest Grundfos sales company.

You can change the settings by means of the following user interfaces:

- HMI 200 operating panel
- HMI 201 operating panel [7)]
- HMI 300 operating panel
- HMI 301 operating panel [7)]
- Grundfos GO application.

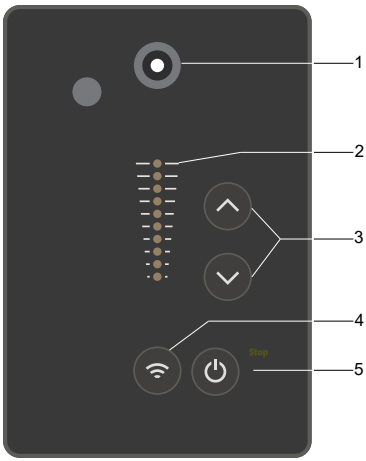
7) HMI without a radio module.

All settings are saved if the power supply is switched off.

Related information

2.4.4 Identification of the operating panel

7.1.1 Operating panels, HMI 200 and HMI 201



Pos.	Symbol	Description
1		**Grundfos Eye:** The indicator light shows the operating status of the product.
2	-	Light fields for indication of the setpoint.
3		**Up**/**Down**: The buttons change the setpoint.
4		**Communication:** The button enables communication with Grundfos GO and other products of the same type.
5		**Start/Stop**: Press the button to make the product mandatory-ready for operation or to start and stop the product. **Start**: If you press the button when the product is stopped, the product starts if no other functions with higher priority have been enabled. **Stop**: If you press the button when the product is running, the product always stops. When you press the button, the stop icon appears at the bottom of the display.

7.1.1.1 Setting the setpoint in constant parameter mode

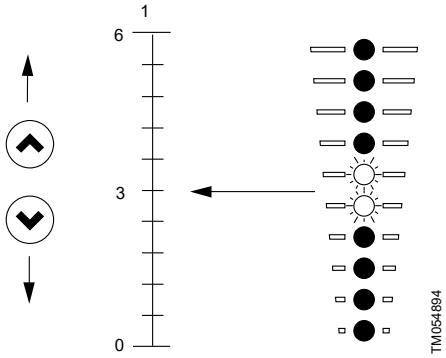
The following applies for motors set to operate in **Const. other val.**

Set the desired setpoint by pressing the **Up** or **Down** buttons.

The green light fields on the operating panel indicate the setpoint set.

The following example applies to a pump or motor in an application where a pressure sensor gives a feedback to the pump or motor. The sensor has been set manually, and the pump or motor does not automatically register a connected sensor.

Light fields 5 and 6 are activated, indicating a desired setpoint of 3 bar with a sensor measuring range from 0 to 6 bar. The setting range is equal to the sensor measuring range.

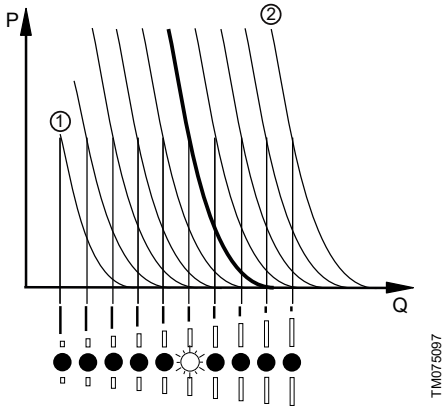


7.1.1.2 Setting the setpoint in constant curve mode

Set the desired setpoint by pressing the **Up** or **Down** buttons.

The green light fields on the operating panel indicate the setpoint set.

Example: In **Constant curve** mode, the motor output is between minimum and maximum speed defined by **Operating range**.

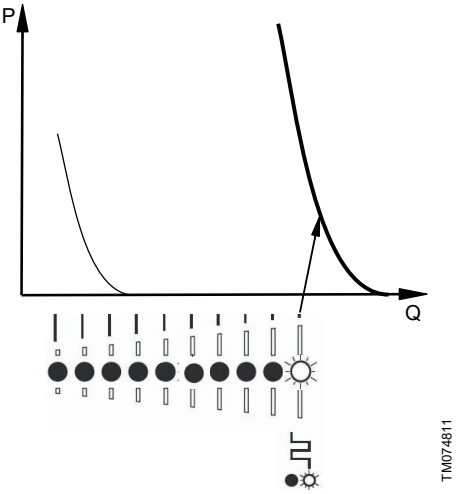


Pos.	Description
p	Pressure
Q	Flow rate
1	Minimum
2	Maximum

7.1.1.3 Setting to maximum speed

The motor must not be in operating mode **Stop**.

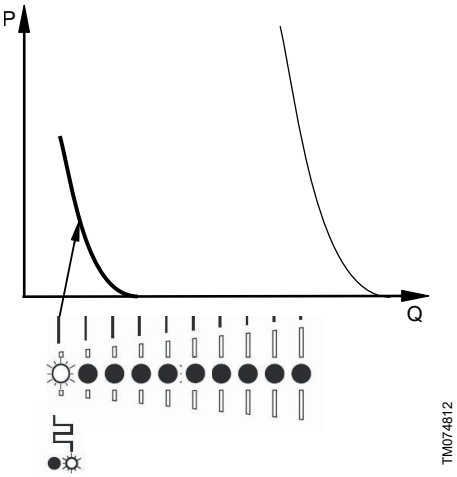
- Press and hold the **Up** button until the top light field is on and starts flashing.



7.1.1.4 Setting to minimum speed

The motor must not be in operating mode **Stop**.

- Press and hold the **Down** button until the bottom light field is on and starts flashing.



7.1.1.5 Starting the pump

How you start the pump depends on how it was stopped.

Start the pump in one of the following ways:

- If the pump was stopped by pressing the **Start/Stop** button, start the pump by pressing the **Start/Stop** button.
- If the pump was stopped by pressing and holding the **Down** button, start the pump by pressing and holding the **Up** button.

7.1.1.6 Stopping the pump

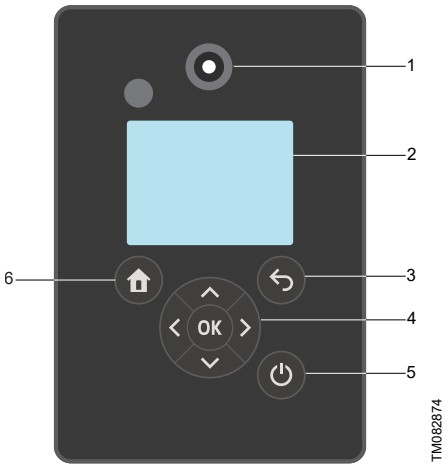
Stop the pump in one of the following ways:

- Press the **Start/Stop** button.
- Press and hold the **Down** button until all light fields are off.
- Use the Grundfos GO.
- Use a digital input set to **External stop**.

7.1.1.7 Resetting alarms and warnings in products with the HMI 200 or HMI 201 operating panel

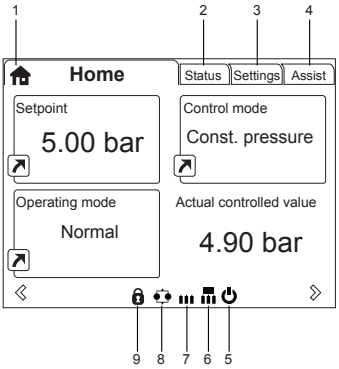
- You can reset a fault indication in one of the following ways:
  - Briefly press the **Up** or **Down** button.
    This is not possible if the buttons have been locked.
    This does not change the setting of the motor.
  - Switch off the power supply until the indicator lights are off.
  - Switch the external start and stop input off, and then on again.
  - Use Grundfos GO.
  - Use the digital input if you have set it to **Alarm resetting**.

7.1.2 Operating panels, HMI 300 and HMI 301



Pos.	Symbol	Description
1		**Grundfos Eye:** The indicator light shows the operating status of the product.
2	-	Graphical colour display.
3		**Back**: Press the button to go one step back.
		Left/**Right**: Press the buttons to navigate between main menus, displays and digits. When you change the menu, the display shows the top display of the new menu.
		Up/**Down**: Press the buttons to navigate between submenus or change the value settings. If you have disabled the possibility to make settings with the **Enable/ disable settings** function, you can enable it again temporarily by pressing these buttons simultaneously for at least 5 seconds.
4	OK	**OK**: Press the button to do as follows: • save changed values, reset alarms and expand the value field • enable communication with Grundfos GO and other products of the same type. When you try to establish radio communication between the product and Grundfos GO or another product, the green indicator light in Grundfos Eye flashes. In the controller display, a note states that a device wants to connect to the product. Press **OK** on the product operating panel to allow communication with Grundfos GO or Grundfos GO Link and other products of the same type.
5		**Start/Stop**: Press the button to make the product mandatory-ready for operation or to start and stop the product. **Start**: If you press the button when the product is stopped, the product starts if no other functions with higher priority have been enabled. **Stop**: If you press the button when the product is running, the product always stops. When you press the button, the stop icon appears at the bottom of the display.
6		**Home**: Press the button to go to the **Home** menu.

7.1.2.1 Home display



Pos.	Symbol	Description
1		**Home**: This menu shows up to four user-defined parameters. You can access each parameter directly from this menu.
2	-	**Status**: This menu shows the status of the product and system, warnings and alarms.
3	-	**Settings**: This menu gives access to all setting parameters. The menu also allows you to make detailed settings.
4	-	**Assist**: This menu enables assisted setup, provides a short description of the control modes and offers fault-finding advice.
5		**Start/Stop**: The icon indicates that the product was stopped with the **Start/Stop** button.
6		**Master:** The icon indicates that the product is functioning as the master in a multipump system.
7		**Slave:** The icon indicates that the product is functioning as a slave in a multipump system.
8		**Multioperation:** The icon indicates that the product is operating in a multipump system.
9		**Lock:** The icon indicates that the possibility to make settings has been disabled for protective reasons.

7.1.2.2 Startup guide

The function is only available in the HMI 300 and HMI 301 operating panels.

The startup guide starts at the first startup and guides you through the settings needed for the product to operate in the given application. When the startup guide has been completed, the main menus appear in the display.

You can always run the startup guide at a later time.

7.1.2.3 Menu overview for the HMI 300 and HMI 301 operating panels

Home	Single pump	Multipump system
	•	•

Status		Single pump	Multipump system
Operating status		•	•
	Operating mode, from	•	•
	Control mode	•	•
Pump performance		•	•
	Actual control. value	•	•
	Resulting setpoint	•	•
	Speed	•	•
	Acc. flow and specific energy	•	•
Power and energy consumption		•	•
Measured values		•	•
	Analog input 1	•	•
	Analog input 2	•	•
	Analog input 3 [8)]	•	•
	Grundfos Direct Sensor	•	•
	Pt100/1000 input 1 [8)]	•	•
	Pt100/1000 input 2 [8)]	•	•
Analog output [8)]		•	•
Warning and alarm		•	•
	Actual warning or alarm	•	•
	Warning log	•	•
	Alarm log	•	•
Operating log		•	•
	Operating hours	•	•
Fitted modules		•	•
Date and time [8)]		•	•
Product identification		•	•
Motor bearing monitoring		•	•
Multi-pump system			•
	System operating status		•
	System performance		•
	System input power and energy		•
	Pump 1, multi-pump system		•

Status		Single pump	Multipump system
	Pump 2, multi-pump system		•
	Pump 3, multi-pump system		•
	Pump 4, multi-pump system		•

8) Only available if an advanced functional module, type FM310 or FM311, is fitted.

Settings		Single pump	Multipump system
Setpoint		•	•
Operating mode		•	•
Set user-defined speed		•	•
Set manual speed		•	•
Control mode		•	•
Setting the proportional pressure		•	
Analog inputs		•	•
	Analog input 1, setup	•	•
	Analog input 2, setup	•	•
	Analog input 3, setup 9)	•	•
	Grundfos Direct Sensor	•	•
Pt100/1000 inputs 9)		•	•
	Pt100/1000 input 1, setup 9)	•	•
	Pt100/1000 input 2, setup 9)	•	•
Digital inputs		•	•
	Digital input 1, setup	•	•
	Digital input 2, setup 9)	•	•
Digital inputs/outputs		•	•
	Digital input/output 3, setup	•	•
	Digital input/output 4, setup 9)	•	•
Relay outputs		•	•
	Relay output 1	•	•
	Relay output 2	•	•
Analog output 9)		•	•
	Output signal 9)	•	•
	Function of analog output 9)	•	•
Controller settings		•	•
Operating range		•	•
Setpoint influence		•	•
	Ext. setpoint infl.	•	•
	Predefined setpoints 9)	•	•

Settings		Single pump	Multipump system
Monitoring functions		•	•
	Motor bearing monitoring	•	•
	Alarm handling	•	•
	Motor bearing maintenance	•	•
	Limit-exceeded function	•	•
	LiqTec function	•	•
Special functions		•	•
	Low-flow stop function	•	•
	Stop at min. speed	•	•
	Pipe filling function	•	•
	Pulse flowmeter setup	•	•
	Ramps	•	•
	Standstill heating	•	•
Communication		•	•
	Pump number	•	•
	Enable/disable radio comm.	•	•
	Enable/disable Bluetooth comm.	•	•
	Initiate Bluetooth connection	•	•
	Setup of AYB terminals	•	•
	Setup of Ethernet	•	•
General settings		•	•
	Language	•	•
	Set date and time	•	•
	Units	•	•
	Enable/disable settings	•	•
	Delete history	•	•
	Define Home display	•	•
	Display settings	•	•
	Store actual settings	•	•
	Recall stored settings	•	•
	Run start-up guide	•	•

9) Only available if an advanced functional module, type FM310 or FM311, is fitted.

Assist	Single pump	Multipump system
Assisted pump setup	•	•
Setup, analog input	•	•
Setting of date and time	•	•

Assist	Single pump	Multipump system
Setup of multi-pump system	•	•
Description of control mode	•	•
Assisted fault advice	•	•

7.2 Grundfos GO

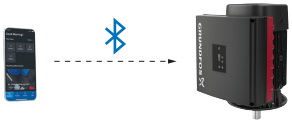
CAUTION
Radiation
Minor or moderate personal injury

- Locate the product at a minimum distance of 20 cm from any body parts. Human tissue may be heated by RF energy.

Installers and end users must be provided with these installation and operating instructions and operating conditions for satisfying RF exposure compliance.

The product is designed for wireless communication with Grundfos GO using Bluetooth (BLE).

Grundfos GO enables you to set functions and gives you access to status overviews, technical product information and current operating parameters.



TM087682

7.2.1 Communication

When Grundfos GO initiates communication with the product, the indicator light in the centre of Grundfos Eye flashes green.

On products fitted with the HMI 100 or HMI 200 operating panel, you can enable communication by pressing the **Communication** button.

On products fitted with the HMI 300 operating panel, the display indicates that a wireless device is trying to connect to the product. Press **OK** on the operating panel to connect the product with Grundfos GO, or press the **Home** button to reject connection.

Symbol	Description
OK	Press **OK** on the operating panel to connect the product with Grundfos GO.
	Press the **Home** button to reject connection.

7.2.1.1 Bluetooth communication

Bluetooth communication can take place at distances up to 10 m. The first time Grundfos GO communicates with the product, you enable communication by pressing the **Communication** button or **OK** on the operating panel.

Later when communication takes place, the product is recognised by Grundfos GO, and you can select the product from the **List** menu.

7.2.2 Menu overview for Grundfos GO

Dashboard	Single pump	Multipump system
	•	•

View all metrics		Single pump	Multipump system
Pump and application			
	Actual controlled value	•	•
	Acc. flow, specific energy	•	•
	Energy consumption	•	
	Energy consumption, system		•
	Power consumption	•	•
	Power consumption, system		•
	Motor bearing service	•	•
	Resulting setpoint	•	
	Resulting system setpoint		•
	Motor speed	•	•
	Pump 1		•
	Pump 2		•
	Pump 3		•
	Pump 4		•
Operating Log			
	Operating hours	•	
	Operating hours, system		•
	Motor current	•	•
	Number of starts	•	
Inputs/outputs			
	Analog input 1	•	
	Analog input 2	•	
	Analog input 3 [10)]	•	
	Grundfos Direct Sensor	•	
	Analog, Output [10)]	•	
	Pt100/1000 input 1 [10)]	•	
	Pt100/1000 input 2 [10)]	•	
	Digital input 1	•	
	Digital input 2 [10)]	•	
	Digital input/output 3	•	
	Digital input/output 4 [10)]	•	
Monitored metrics			

View all metrics		Single pump	Multipump system
	Ambient temperature	•	•
	Differential pressure	•	•
	Differential pressure, inlet/outlet	•	•
	Differential temperature, external	•	•
	External pressure 1	•	•
	External pressure 2	•	•
	Feed tank pressure	•	•
	Flow rate	•	•
	Pressure : inlet	•	•
	Pressure : outlet	•	•
	Other parameter	•	•
	Tank pressure, external	•	•
	Temperature 1	•	•
	Temperature 2	•	•
	Conductivity	•	•
Fitted modules			
	Functional module	•	
	Power board	•	
	CIM module	•	
	Operating panel	•	

10) Only available if an advanced functional module, type FM310 or FM311, is fitted.

Settings		Single pump	Multipump system
Pump and application			
	Pump name	•	•
	Control mode	•	•
	Operating mode	•	•
	Setpoint	•	•
	Set user-defined speed	•	•
	Operating range	•	•
	Controller	•	•
	External setpoint funct.	•	•
	Predefined setpoint	•	•
	Setting the proportional pressure	•	•
	FLOWLIMIT	•	
	Lock panel	•	
	Service	•	
	Alternating operation, time		•

Settings		Single pump	Multipump system
	Sensor to be used		•
	Time for pump changeover [11)]		•
Inputs/outputs			
	Analog input 1	•	
	Analog input 2	•	
	Analog input 3 [11)]	•	
	Grundfos Direct Sensor	•	
	Analog output [11)]	•	
	Pt100/1000 input 1 [11)]	•	
	Pt100/1000 input 2 [11)]	•	
	Digital input 1	•	
	Digital input 2 [11)]	•	
	Digital input/output 3	•	
	Digital input/output 4 [11)]	•	
	Relay output 1	•	
	Relay output 2	•	
Monitoring functions			
	Alarm handling	•	
	Limit 1 exceeded	•	
	Limit 2 exceeded	•	
	Limit 3 exceeded	•	
	Limit 4 exceeded	•	
	LiqTec function	•	
	Motor bearing monitoring	•	
Special functions			
	Low-flow stop	•	•
	Pipe-filling function	•	•
	Pulse flow meter	•	•
	Ramps	•	•
	Standstill heating	•	•
	Stop at min. speed	•	•
Communication			
	Bluetooth communication	•	•
	Radio communication	•	•
	GENIbus Number	•	
	Connectivity and port settings	•	
General			
	Connection code	•	

Settings		Single pump	Multipump system
	Date and time [11)]	•	
	Firmware	•	
	Store settings	•	
	Recall settings	•	
	Unit configuration	•	

11) Only available if an advanced functional module, type FM310 or FM311, is fitted.

Alarms and warnings	Single pump	Multipump system
Alarm log	•	•
Warning log	•	•

Setup	Single pump	Multipump system
Assisted pump setup	•	
Assisted fault advice	•	•
Application wizard	•	
Multi-pump setup	•	•

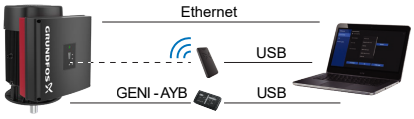
7.3 Grundfos GO Link

The product is designed for wired or wireless communication with the Grundfos GO Link.

The Grundfos GO Link enables you to set functions and gives you access to status overviews, configuration and current operating parameters.

Use the Grundfos GO Link together with the following interfaces:

- Ethernet cable (Only FM310 and FM311)
- Grundfos MI 301 - USB - Wired/wireless (Only HMI 100, HMI 200 and HMI 300)
- Grundfos PC Tool Link - USB - Wired.



Grundfos GO Link setup

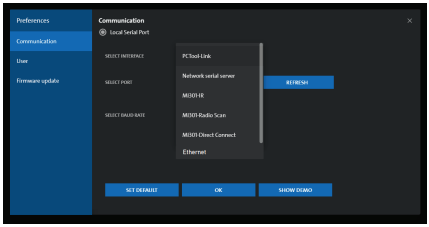
Pos.	Description
1	Ethernet cable: Standard Ethernet cable CAT5/CAT6
2	Grundfos MI 301: Separate radio equipment enabling radio communication [12)]
3	Grundfos PC Tool Link: Separate module enabling wired connection to the pump [12)]

12) Use the module together with a USB cable to connect to a laptop.

7.3.1 Communication

When Grundfos GO Link initiates communication with the product, it is done using different verification methods.

Select the interface connected to the pump:



7.3.2 Ethernet

Wired connection can take place using an Ethernet cable connected directly between a laptop and the RJ45 interface in the pump or via a local network having both the pump and the laptop connected to the same network.

To establish a secure connection between the laptop and the pump, the user will have to go through a verification process.

Connecting to a pump can either happen by scanning for a connected product, which can be a direct Ethernet connection, or the pump is connected to a local network or a connection via the pump IP address.

Initiate connection from Grundfos GO Link and follow onscreen instructions.

7.3.3 Grundfos MI 301

Radio communication can take place at distances up to 30 metres. The first time Grundfos GO Link communicates with the product, you enable communication by pressing the **Radio communication** button or **OK** on the operating panel. Select either MI301-Direct connect or MI301-Radio. When communication takes place, the product is recognized by Grundfos GO Link, and you can connect using Direct connect or Radio scan without having to run a verification.

7.3.4 Grundfos PC Tool Link

Wired connection can take place using Grundfos PC Tool connected to the AYB terminals of the pump. Since Grundfos GO Link is wired to the pump within a short distance, no verification is needed. A direct connection will be established.

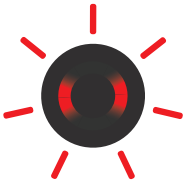
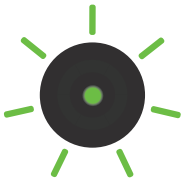
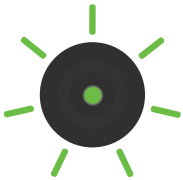
7.4 Grundfos Eye

The operating condition of the motor is indicated by Grundfos Eye (A) on the motor operating panel.



Grundfos Eye indicator light

Indicator light	Indication	Description
	No lights are on.	**Power off** The motor is not running.
	Two opposite green indicator lights are rotating.	**Power on** The motor is running. The indicator lights are rotating in the direction of rotation of the motor when seen from the non-drive end.
	Two opposite green indicator lights are permanently on.	**Power on** The motor is not running.
	One yellow indicator light is rotating.	**Warning** The motor is running. The indicator light is rotating in the direction of rotation of the motor when seen from the non-drive end.
	One yellow indicator light is permanently on.	**Warning** The motor has stopped.

Indicator light	Indication	Description
	Two opposite red indicator lights are flashing simultaneously.	**Alarm** The motor has stopped.
	The green indicator light in the middle flashes quickly four times.	Grundfos Eye flashes four times when you press the Grundfos Eye symbol next to the motor name in Grundfos GO.
	The green indicator light in the middle is flashing continuously.	You have selected the motor in Grundfos GO, and the motor is mandatory-ready to be connected.
	The green indicator light in the middle flashes quickly for a few seconds.	The motor is controlled by Grundfos GO or exchanging data with Grundfos GO.
	The green indicator light in the middle is permanently on.	The motor is connected to Grundfos GO.

Related information

11. Fault finding

8. Setting the product

You can set control functions via Grundfos GO, Grundfos GO Link or the HMI 300 or HMI 301 operating panel.

- If only one function name is mentioned, it refers to both Grundfos GO and the operating panel.
- If a function name is mentioned in a parenthesis, it refers to a function on the operating panel.

Related information

8.5 Control mode

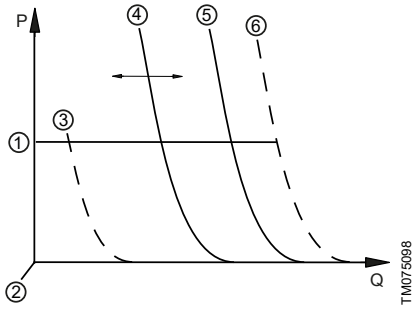
8.1 Setpoint

When you have selected the desired control mode, set the setpoint.

8.2 Operating mode

Possible operating modes

Normal	The product runs according to the selected control mode.
Stop	The product stops.
Min.	The product runs at minimum speed. You can use the minimum curve mode in periods in which a minimum flow is required. When operating according to the minimum curve, the pump is operating like an uncontrolled pump.
Max.	The product runs at maximum speed. You can use the maximum curve mode in periods in which a maximum flow is required. When operating according to the maximum curve, the pump is operating like an uncontrolled pump.
Manual	The product is operating at a manually set speed, and the setpoint via bus and setpoint influence function are overruled.
User-defined speed	The product is operating at a speed set by the user.



Pos.	Description
p	Pressure
Q	Flow rate
1	Normal
2	Stop
3	Minimum
4	Manual
5	Normal
6	Maximum

8.3 Set manual speed

The function is only available in the HMI 300 and HMI 301 operating panels.

Use this function to set the speed in percentage of the maximum speed. When you have set the operating mode to **Manual**, the product starts running at the set speed.

With Grundfos GO, you can set the speed via the **Setpoint** menu.

8.4 Set user-defined speed

Use this function to set the motor speed in percentage of the maximum speed. When you set the operating mode to **User-defined speed**, the motor starts running at the set speed.

8.5 Control mode

You can choose between the following control modes:

- **Constant pressure**
- **Const. other val.** (constant other value)
- **Const. curve** (constant curve)

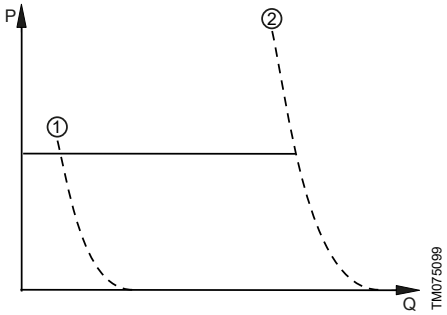
Related information

8. Setting the product

8.5.1 Constant pressure

Use this control mode if you want the pump to deliver a constant pressure. The control mode uses the factory-fitted pressure sensor, if any, which measures the outlet pressure of the pump.

For pumps without a factory-fitted sensor, you must connect a pressure sensor to one of the analog inputs of the pump. You can set the pressure sensor in the **Assist** menu.



Pos.	Description
P	Pressure
Q	Flow rate
1	Minimum
2	Maximum

Related information

8.13 Controller settings

8.41 Assisted pump setup

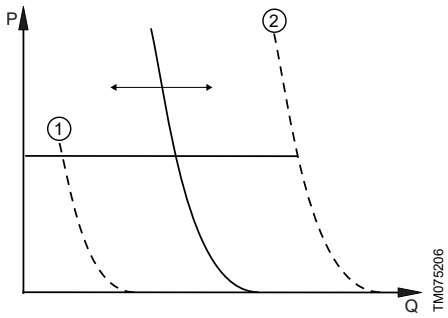
8.5.2 Constant other value

Use this control mode to control a value not available in the **Control mode** menu. To measure the controlled value, connect a sensor to one of the analog inputs. The controlled value is shown in percentage of the sensor range.

8.5.3 Constant curve

Use this control mode to control the motor speed.

You can set the desired speed in percentage of the maximum speed in the range from user-set minimum speed to user-set maximum speed.



Pos.	Description
P	Pressure
Q	Flow rate
1	Minimum
2	Maximum

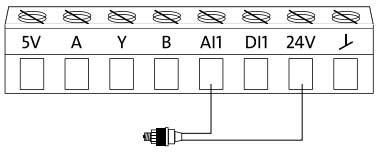
8.6 Analog inputs

The inputs and outputs available depend on the functional module fitted in the motor.

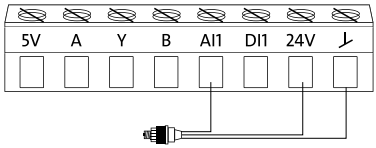
Functional module	Analog input 1 (Terminal AI1)	Analog input 2 (Terminal AI2)	Analog input 3 (Terminal AI3)
FM300	•	•	•
FM310	•	•	•
FM311	•	•	•

Wiring examples:

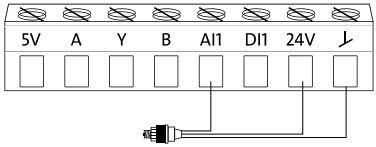
These connection scenarios are also valid for connection to analog input 2 and analog input 3.



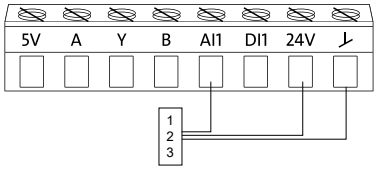
2-wire sensor, 0/4-20 mA



3-wire sensor, 0/4-20 mA



3-wire sensor, 0.5 - 3.5 V, 0-5 V, 0-10 V



Setpoint influence, 0.5 - 3.5 V, 0-5 V, 0-10 V; 0/4-20 mA

Pos.	Description
1	Potentiometer
2	PLC
3	External controller

To set the input, make the settings below:

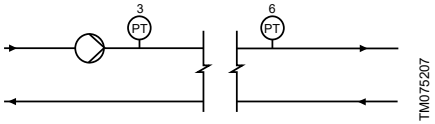
Function

You can set the inputs to these functions:

- **Not active**
- **Feedback sensor**
  The sensor is used for the selected control mode.
- **Setpoint influence**
  The input signal is used for influencing the setpoint.
- **Other function**
  The sensor input is used for measurement or monitoring.

Measured parameter

Select one of the below parameters to be measured in the system by the sensor connected to the input.



Pos.	Sensor function/measured parameter
3	**Outlet pressure**
6	**Press. 1, external**

Unit

Parameter	Available units
Pressure	bar, m, kPa, psi, ft
Level	cm, m, ft, in
Pump flow	m^3/h, l/s, yd^3/h, gpm
Liquid temperature	°C, °F
Other parameter	%

Electrical signal

Available signal types:

- 0.5 - 3.5 V
- 0-5 V
- 0-10 V
- 0-20 mA
- 4-20 mA.

Sensor range, minimum value

Set the minimum value of the connected sensor.

Sensor range, maximum value

Set the maximum value of the connected sensor.

8.6.1 Setting two sensors for differential measurement

Two analog sensors must be installed and connected electrically to measure a parameter at two different locations in a system.

The pressure, temperature and flow parameters can be used for differential measurement.

- Set the analog inputs according to the measured parameter:

Parameter	Sensor 1, measured parameter	Sensor 2, measured parameter
Pressure, option 1	**Inlet pressure**	**Outlet pressure**
Pressure, option 2	**Press. 1, external**	**Press. 2, external**
Flow	**Pump flow**	**Flow, external**
Temperature	**Temperature 1**	**Temperature 2**

If you want to use the **Con. diff. press.**, **Con. diff. temp.** or **Const. flow rate** control modes, you must configure both sensors as **Feedback sensor**.

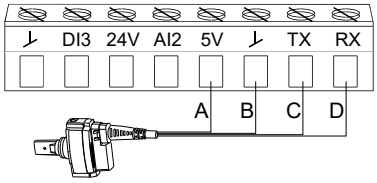
8.7 Grundfos Direct Sensor

Grundfos Direct Sensor is a digital sensor that auto detects range and unit.

Grundfos Direct Sensor always has the capability to also measure the media temperature. The pump will automatically detect range and unit of the temperature sensor.

For information about the functions and measured parameters of each sensor, see the sections on the sensor, temperature and dry-running protection.

Wiring example:



TM086416

Designation	Colour
A	Brown
B	Green
C	Yellow
D	White

Related information

8.41 Assisted pump setup

8.47 Factory settings for Grundfos GO

8.7.1 Sensor

Function

You can set the sensor to the following functions:

- **Not active**
- **Feedback sensor**

  The sensor is used for the selected control mode.
- **Setpoint influence**

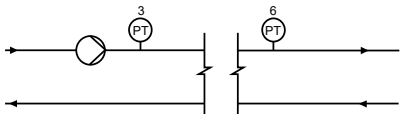
  The input signal is used for influencing the setpoint.
- **Other function**

  The sensor input is used for measurement or monitoring.

Measured parameter

Select one of the below parameters to be measured in the system by the sensor connected to the input.

Note that the list will be reduced to match the installed sensor.



TM075207

Pos.	Sensor function/ measured parameter
3	**Outlet pressure**
6	**Press. 1, external**

8.7.2 Temperature

Function

You can set the sensor to the following functions:

- **Not active**
- **Feedback sensor**

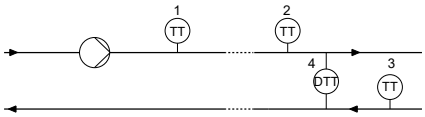
  The sensor is used for the selected control mode.
- **Setpoint influence**

  The input signal is used for influencing the setpoint.
- **Other function**

  The sensor input is used for measurement or monitoring.

Measured parameter

Select one of the below parameters to be measured in the system by the sensor connected to the input.



TM064012

Pos.	Sensor function/ measured parameter
1	**Liquid temp.**
2	**Temperature 1**
3	**Temperature 2**
4	**Differential temp.**
Not shown	**Ambient temp.**

8.7.3 Dry-running protection

Use this function to set dry-running protection to **Enabled** or **Disabled**.

The function requires that a CPS sensor has been fitted in the pump head and connected to the pump. When you have enabled the dry-running protection function, it stops the pump if dry running occurs. Restart the pump manually if it has been stopped due to dry running.

Dry-running detection delay

You can set a detection delay to make sure that the pump is given a chance to start up and pump the air in the pump out before the dry-running protection function again detects dry running and stops the pump.

More than 10 seconds of dry running can damage the shaft seal and can reduce the lifetime of the product.

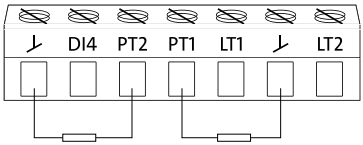
Range: 0-254 seconds.

8.8 Pt100/1000 inputs

The inputs and outputs available depend on the functional module fitted in the motor.

Functional module	Pt100/1000 input 1 (Terminals PT1, GND)	Pt100/1000 input 2 (Terminals PT2, GND)
FM300	•	•
FM310	•	•
FM311	•	•

Wiring example:



TM083189

Pt100/1000

To set the input, choose one of the below settings.

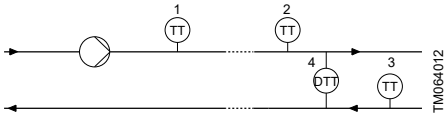
Function

You can set the inputs to these functions:

- **Not active**
- **Feedback sensor**
  The sensor is used for the selected control mode.
- **Setpoint influence**
  The input signal is used for influencing the setpoint.
- **Other function**
  The sensor input is used for measurement or monitoring.

Measured parameter

Select one of the below parameters to be measured in the system by the sensor connected to the input.



TM064012

Pos.	Sensor function/measured parameter
1	**Liquid temp.**
2	**Temperature 1**
3	**Temperature 2**
4	**Differential temp.**
Not shown	**Ambient temp.**

Measuring range

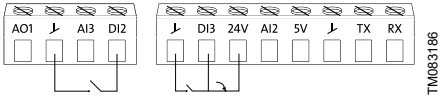
-50 to +204 °C.

8.9 Digital inputs

The inputs and outputs available depend on the functional module fitted in the motor.

Functional module	Digital input 1 (Terminals DI1, GND)	Digital input 2 (Terminals DI2, GND)
FM300	•	•
FM310	•	•
FM311	•	•

Wiring example:



Digital input

To set the input, make the settings below:Function You can set the inputs to these functions:

- **Not active**

  When set to **Not active**, the input has no function.
- **Ext. stop**

  When the input is deactivated, open circuit, the motor stops.
- **Min.** (minimum speed)

  When the input is activated, the motor runs at the set minimum speed.
- **Max.** (maximum speed)

  When the input is activated, the motor runs at the set maximum speed.
- **User defined speed**

  When the input is activated, the motor runs at a speed set by the user.
- **External fault**

  When the input is activated, a timer is started. If the input is activated for more than 5 seconds, the motor stops and a fault is indicated. The function depends on input from external equipment.
- **Alarm resetting**

  When the input is activated, a fault indication, if any, is reset.
- **Dry running**

  When this function is selected, a lack of inlet pressure or water shortage (dry running) can be detected. When this happens, the pump stops. The pump cannot restart as long as the input is activated. This requires the use of an accessory such as these:
  - a pressure switch installed on the inlet side of the pump
  - a float switch installed on the inlet side of the pump.
- **Accumulated flow**

  When this function is selected, the accumulated flow can be registered. This requires the use of a flowmeter which can give a feedback signal as a pulse per defined volume of water.
- **Reverse rotation**

  This function reverses the direction of rotation of the motor.
- **Predefined setpoint 1**

  The function applies only to digital input 2.

  When you set digital inputs to a predefined setpoint, the pump operates according to a setpoint based on a combination of the activated digital inputs.
- **Activate output**

  When this function is selected, the related digital output is activated. This is done without any changes to pump operation.
- **Local motor stop**

  When the function is selected, the given motor in a multimotor system setup stops without affecting the performance of the other motors in the system.

The priority of the selected functions are interdependent.A stop command always has the highest priority.Activation of digital inputsYou can set the digital inputs to trigger on either Closed contact or Open contact. Selecting the trigger function can only be set via Grundfos GO Link.The digital inputs can be activated either as active low or active high.The digital inputs will react as described in the table below:

Activate/ Closed contact	Deactivate/Open contact
GND/0V	Floating/3-24V

8.9.1 Timer function for a digital input

Activation delay

The activation delay (T1) is the time between the digital signal and the activation of the selected function.

Range: 0-6000 seconds.

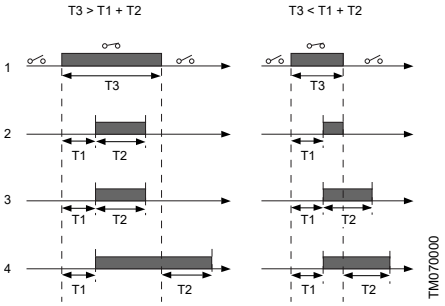
Duration time

Available modes:

- **Not active**
- **Active with interrupt**
- **Active without interrupt**
- **Active with after-run**.

The duration time (T2) is the time which, together with the mode, determines how long the selected function is active.

Range: 0 - 15,000 seconds.



Pos.	Description
1	**Digital input**.
2	**Active with interrupt**.
3	**Active without interrupt**.
4	**Active with after-run**.
T1	**Activation delay**
T2	**Duration time**
T3	The period of time when the digital input is activated.

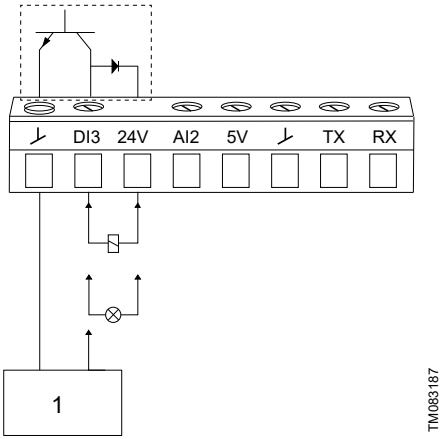
8.10 Digital inputs/outputs

The inputs and outputs available depend on the functional module fitted in the motor.

Functional module	Digital input/ output 3 (Terminals DI3, GND)	Digital input/ output 4 (Terminals DI4, GND)
FM300	•	•
FM310	•	•
FM311	•	•

You can select whether the interface is to be used as an input or output. The output is an open collector. You can connect the open collector to, for example, an external relay or a controller such as a PLC.

Wiring example:



Digital output, open collector

Pos.	Description
1	External controller

Mode

You can set the digital input or output 3 and 4 to act as a digital input or digital output.

Functions if the digital input or output is set to input:

- **Not active**
- **Ext. stop**
- **Min.**
- **Max.**
- **User defined speed**
- **External fault**
- **Alarm resetting**
- **Dry running**
- **Accumulated flow**
- **Predefined setpoint 2** (digital input/output 3)
- **Predefined setpoint 3** (digital input/output 4)
- **Local motor stop**
- **Activate output**

Functions if the digital input or output is set to output:

- **Not active**
- **Ready**
- **Alarm**
- **Operation**
- **Pump running**
- **Warning**
- **Limit 1 exceeded**
- **Limit 2 exceeded**

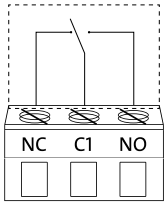
- **Limit 3 exceeded**
- **Limit 4 exceeded**
- **Digital input 1, state**
- **Digital input 2, state**
- **Digital input 3, state**
- **Digital input 4, state**

8.11 Relay outputs

The motor has two outputs for potential-free signals via two internal relays.

Functional module	Signal relay 1 (Terminals NC, C1, NO)	Signal relay 2 (Terminals NC, C2, NO)
FM300	•	•
FM310	•	•
FM311	•	•

Wiring example:



TM083188

Relay output

FunctionsYou can configure the signal relays to be activated when the product changes to one of the following states:

- **Not active**

  The relay has been deactivated.
- **Ready**

  The motor may be running or is ready to run, and no alarms are active.
- **Alarm**

  There is an active alarm, and the motor is stopped.
- **Operating** (**Operation**)

  Operating equals **Running**, but the motor is still in operation when it is stopped, for example, by the **Stop function** or **Limit exceeded**.
- **Running** (**Pump running**)

  The motor shaft is rotating.
- **Warning**

  There is an active warning.
- **Limit 1 exceeded**

  When you have set this function and the limit is exceeded, the signal relay is activated.
- **Limit 2 exceeded**

  When you have set this function and the limit is exceeded, the signal relay is activated.
- **Limit 3 exceeded**

  When you have set this function and the limit is exceeded, the signal relay is activated.
- **Limit 4 exceeded**

  When you have set this function and the limit is exceeded, the signal relay is activated.
- **External fan control** (**Control of external fan**)

  When you select this function, the relay is activated if the internal temperature of the motor electronics reaches a preset limit value. In this way the relay activates external cooling to add additional cooling to the motor.
- **Digital input 1, state**

  Follow digital input 1. If digital input 1 is triggered, the digital output is also triggered.
- **Digital input 2, state**

  Follow digital input 2. If digital input 2 is triggered, the digital output is also triggered.
- **Digital input 3, state**

  Follow digital input 3. If digital input 3 is triggered, the digital output is also triggered.
- **Digital input 4, state**

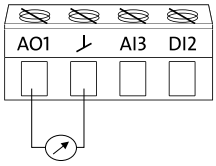
  Follow digital input 4. If digital input 4 is triggered, the digital output is also triggered.

8.12 Analog output

The inputs and outputs available depend on the functional module fitted in the motor.

Functional module	Analog output (Terminals AO, GND)
FM300	•
FM310	•
FM311	•

Wiring example:



TM083185

Analog output, 0/4-20 mA, 0-10 V

The analog output enables external control systems to read specific operating data.

To set the analog output, make the following settings.

Output signal

Possible signal types:

- 0-10 V
- 0-20 mA
- 4-20 mA.

Function of analog output

Actual speed		
0 %	**100 %**	**200 %**
0 V	5 V	10 V
0 mA	10 mA	20 mA
4 mA	12 mA	20 mA

Sensor value	
Minimum	**Maximum**
0 V	10 V
0 mA	20 mA
4 mA	20 mA

Resulting setpoint	
0 %	**100 %**
0 V	10 V
0 mA	20 mA
4 mA	20 mA

Motor load	
0 %	**100 %**
0 V	10 V
0 mA	20 mA
4 mA	20 mA

Motor current		
0 %	**100 %**	**200 %**
0 V	5 V	10 V
0 mA	10 mA	20 mA
4 mA	12 mA	20 mA

Limit-exceeded function	
Output not active	**Output active**
0 V	10 V
0 mA	20 mA
4 mA	20 mA

Flow rate		
0 %	**100 %**	**200 %**
0 V	5 V	10 V
0 mA	10 mA	20 mA
4 mA	12 mA	20 mA

8.13 Controller settings

The pumps have a factory default setting of gain (K_p) and integral time (T_i).

However, if the factory setting is not the optimum setting, you can change the gain and the integral time:

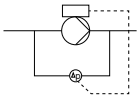
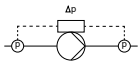
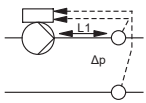
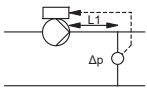
- Set the gain within the range from 0.1 to 20.
- Set the integral-action time within the range from 0.1 to 3600 seconds. If you select 3600 seconds, the controller functions as a PI controller.

Furthermore, you can set the controller to inverse control.

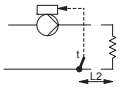
This means that if you increase the setpoint, the speed is reduced. In case of inverse control, you must set the gain within the range from -0.1 to -20.

Guidelines for setting of PI controller

The tables below show the recommended controller settings:

Constant differential pressure	K_p	T_i
	0.5	0.5
		
	0.5	L1 < 5 m: 0.5 L1 > 5 m: 3 L1 > 10 m: 5
		

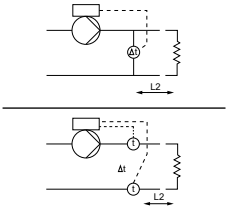
L1: Distance in metres between the pump and the sensor.

Constant temperature	K_p Heating system 13)	K_p Cooling system 14)	T_i
	0.5	-0.5	10 + 5L2
	0.5	-0.5	30 + 5L2

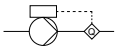
13) In heating systems, an increase in pump performance results in a rise in temperature at the sensor.

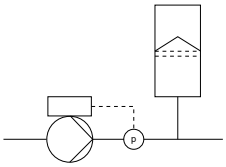
14) In cooling systems, an increase in pump performance results in a drop in temperature at the sensor.

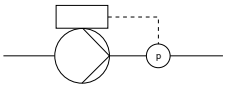
L2: Distance in metres between the heat exchanger and the sensor.

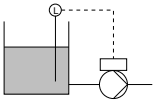
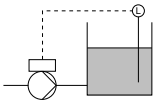
Constant differential temperature	K_p	T_i
	-0.5	10 + 5L2

L2: Distance in metres between the heat exchanger and the sensor.

Constant flow rate	K_p	T_i
	0.5	0.5

Constant pressure	K_p	T_i
	0.5	0.5

Constant pressure	K_p	T_i
	0.5	0.5

Constant level	K_p	T_i
	-10	0
	10	0

General rules of thumb:

If the controller is too slow-reacting, increase the gain.

If the controller is hunting or unstable, dampen the system by reducing the gain or increasing the integral time.

Related information

8.5.1 Constant pressure

8.47 Factory settings for Grundfos GO

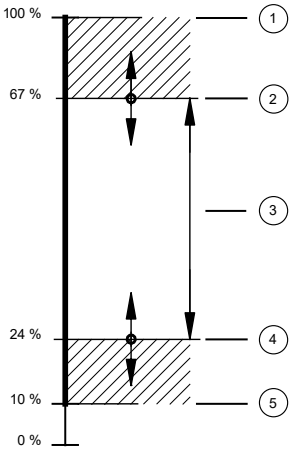
8.14 Operating range

Set the operating range as follows:

1. Set the minimum speed within the range from fixed minimum speed (5) to user-set maximum speed (2).

2. Set the maximum speed within the range from user-set minimum speed (4) to fixed maximum speed (1).

   The range between the user-set minimum and maximum speed is the operating range (3).



Pos.	Description
1	Fixed maximum speed
2	User-set maximum speed
3	Operating range
4	User-set minimum speed
5	Fixed minimum speed

8.15 External setpoint function

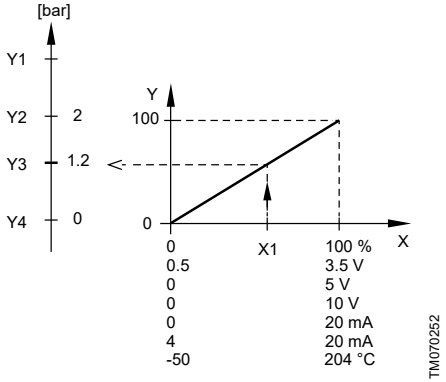
Use this function to influence the setpoint by an external signal via one of the analog inputs.

To enable the function, set one of the analog inputs to **Setpoint influence** with Grundfos GO or to **Ext. setpoint infl.** with the HMI 300 or HMI 301 operating panel.

Example of setpoint influence in control mode Const. pressure

Actual setpoint: actual input signal × setpoint.

At a setpoint of 2 bar and an external setpoint of 60 %, the actual setpoint is 0.60 × 2 = 1.2 bar.

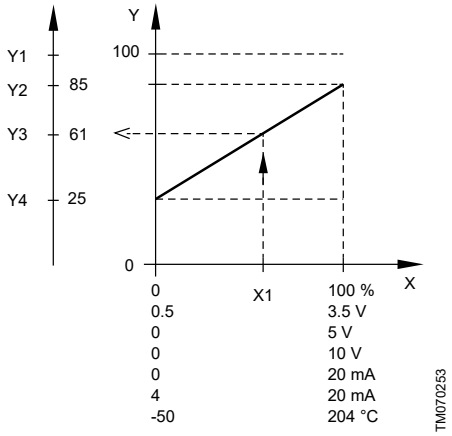


Pos.	Description
X:	External input signal from 0 to 100 %
Y:	Setpoint influence from 0 to 100 %
X1:	Actual input signal, 60 %
Y1:	Sensor maximum
Y2:	Setpoint
Y3:	Actual setpoint
Y4:	Sensor minimum

Example of a constant curve with linear influence function

Actual setpoint: actual input signal × (setpoint - user-set minimum speed) + user-set minimum speed.

At a user-set minimum speed of 25 %, a setpoint of 85 % and an external setpoint of 60 %, the actual setpoint is 0.60 × (85 - 25) + 25 = 61 %.

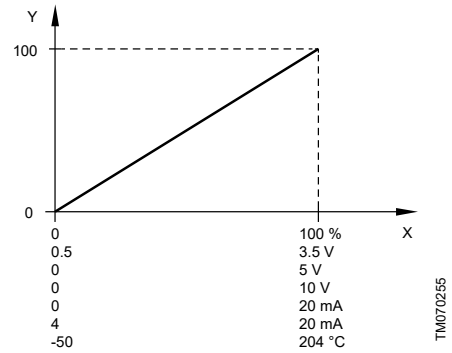


Pos.	Description
X:	External input signal from 0 to 100 %
Y:	Setpoint influence from 0 to 100 %
X1:	Actual input signal, 60 %
Y1:	Fixed maximum speed in percentage
Y2:	Setpoint speed in percentage
Y3:	Actual setpoint speed in percentage
Y4:	User-set minimum speed in percentage

8.15.1 Setpoint influence functions

8.15.1.1 Linear function

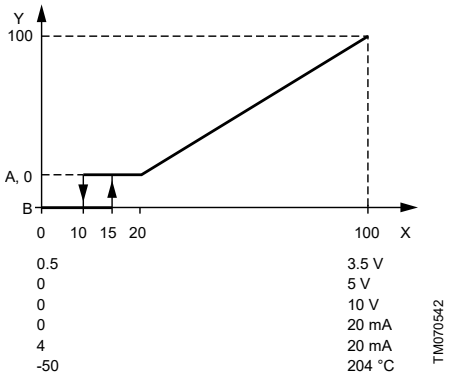
The setpoint is influenced linearly from 0 to 100 %.



Pos.	Description
X	External input signal from 0 to 100 %
Y	Setpoint influence from 0 to 100 %

8.15.1.2 Linear with Stop

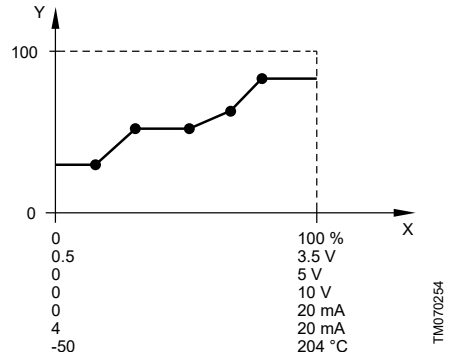
In the input signal range from 20 to 100 %, the setpoint is influenced linearly. If the input signal is below 10 %, the motor changes to the **Stop** operating mode. If the input signal increases over 15 %, the operating mode changes back to **Normal**.



Pos.	Description
X	External input signal from 0 to 100 %
Y	Setpoint influence from 0 to 100 %
A	**Normal**
B	**Stop**

8.15.1.3 Influence table

The setpoint is influenced by a curve made of two to eight points. There is a straight line between the points and a horizontal line before the first point and after the last point.



Pos.	Description
X	External input signal from 0 to 100 %
Y	Setpoint influence from 0 to 100 %

8.16 Predefined setpoints

You can set and activate seven predefined setpoints by combining the input signals with digital inputs 2, 3 and 4, as shown in the table below. Set the digital inputs 2, 3 and 4 to **Predefined setpoints** if all seven predefined setpoints are to be used. You can also set one or two of the digital inputs to **Predefined setpoints**. However, this limits the number of predefined setpoints available.

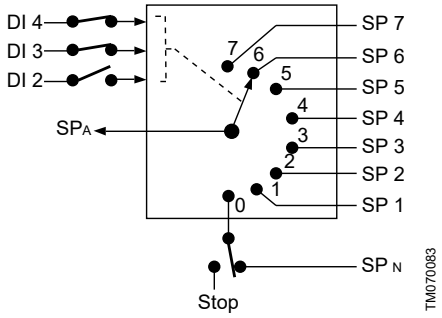
Digital inputs			Setpoint
2	3	4	
0	0	0	Normal setpoint or **Stop**
1	0	0	**Predefined setpoint 1**
0	1	0	**Predefined setpoint 2**
1	1	0	**Predefined setpoint 3**
0	0	1	**Predefined setpoint 4**
1	0	1	**Predefined setpoint 5**
0	1	1	**Predefined setpoint 6**
1	1	1	**Predefined setpoint 7**

0: Open contact

1: Closed contact

Example

The figure shows how you can use the digital inputs to set seven predefined setpoints. Digital input 2 is open, and digital inputs 3 and 4 are closed. If you compare with the table above, you can see that **Predefined setpoint 6** is activated.



Pos.	Description
DI	**Digital input**
SP	**Setpoint**
SP_A	**Actual setpoint**
SP_N	Normal setpoint
Stop	**Stop**

If all digital inputs are open, the motor stops or runs at the normal setpoint. Set the desired action with the Grundfos GO or with the HMI 300 or HMI 301 operating panel.

8.17 Limit-exceeded function

Use this function to monitor a measured parameter or one of the internal values such as speed, motor load or motor current. If a set limit is reached, a selected action can take place. You can set up to four limit-exceeded functions, meaning that you can monitor four different parameters or two to four limits of the same parameter simultaneously.

Note that when using **Limit 1-3 exceeded** in a multipump system, the selected action has an impact on the system, for example, if Action is set to Stop, then the system stops.

Limit 4 exceeded in a multipump system is a local function. The selected action only has an impact on the single pump, for example, if Action is set to Stop, then only the single pump stops.

For **Limit 4 exceeded** in a multipump system, the action always leads to a pump stop, alarm and stop or a warning.

The function requires setting the following parameters:

Measured

Set the measured parameter to be monitored.

Limit

Set the limit that activates the function.

Hysteresis band

Set the hysteresis band for when the function must be deactivated again.

Limit exceeded when

Set the function to be activated when the selected parameter exceeds or drops below the set limit.

- **above limit**:
  The function is activated if the measured parameter exceeds the set limit.
- **below limit**:
  The function is activated if the measured parameter drops below the set limit.

Action

If the value exceeds a limit, you can set an action. The following actions are available:

- **Not active**:

  The pump remains in its current state. Use this setting if you only want to activate a signal relay output when the limit is reached.
- **Stop**:

  The pump stops.
- **Min.**:

  The pump reduces the speed to minimum speed.
- **Max.**:

  The pump increases the speed to maximum speed.
- **User-defined speed**:

  The pump runs at a speed set by the user.
- **Alarm and Stop**:

  An alarm is given, and the pump stops.
- **Alarm and Min.**:

  An alarm is given, and the pump decreases the speed to a minimum.
- **Alarm and Max.**:

  An alarm is given, and the pump increases the speed to maximum.
- **Alarm and User-defined speed**:

  An alarm is given, and the pump runs at the speed set by the user.

Alarm and Warning text

The **Limit-exceeded** function automatically defines a relevant alarm or warning text based on the **Measured parameter** and **Limit exceeded when** functions.

The auto defined text can be overwritten by selecting the alarm or warning text option **Limit X exceeded**.

The following list shows the auto-defined texts:

- Limit X exceeded
- Low inlet pressure
- High discharge pressure
- High pressure
- Low pressure
- High temperature
- Low temperature
- High flow
- Low flow
- High level
- Low level
- High diff. pressure
- Low diff. pressure
- High conductivity.

Detection in Stop

Enable this function to prevent the pump from monitoring the limit while the pump is in the **Stop** state.

Use the **Detection delay** function to allow the pump to start up and bring the value above the limit before detecting.

Detection delay

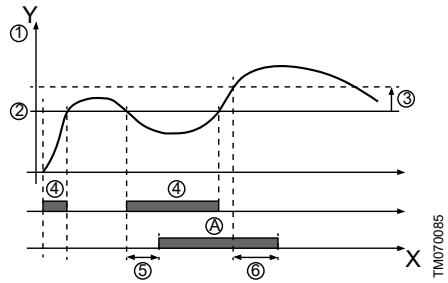
Setting the detection delay ensures that the monitored parameter stays above or below a set limit in a set time before the function is activated.

Resetting delay

The resetting delay is the time from the point when the measured parameter differs from the set limit, including the set hysteresis band, to the point when the function is reset.

Example

The function is to monitor the outlet pressure from the pump. If the pressure is below 5 bar for more than 5 seconds, a warning is indicated. If the pressure is above 7 bar for more than 8 seconds, reset the limit-exceeded warning.



X: Time in seconds

Y: Pressure in bar

Pos.	Parameter	Setting
1	**Measured**	**Discharge pressure**
2	**Limit**	5 bar
3	**Hysteresis band**	2 bar
4	**Limit exceeded when**	**below limit**
5	**Detection delay**	5 seconds
6	**Resetting delay**	8 seconds
A	**Limit-exceeded function active**	-
-	**Action**	**Warning**

8.18 Stop at min. speed

This stop function can be utilised in, for example, constant level applications where a boost of pressure is not needed. This stop function differs from the low-flow stop but the purpose is the same. The pump stops if there is no or low consumption.

This function monitors the speed of the pump. When the PI-controller has forced the speed of the pump to minimum according to the feedback value, the pump stops after a set period of time. It remains stopped until the feedback value starts to drop and the PI-controller starts the pump again.

- **Enable Stop at min. speed**

  Enables the function **Stop at min. speed**.
- **Delay**

  The delay time during which the pump must be running at minimum speed before it stops.
- **Restart speed**

  It is the speed in percentage when the pump must start again, hysteresis. It must be set higher than the minimum speed of the pump.

8.19 Ramps

The ramps determine how quickly the product can accelerate and decelerate during start and stop or setpoint changes.

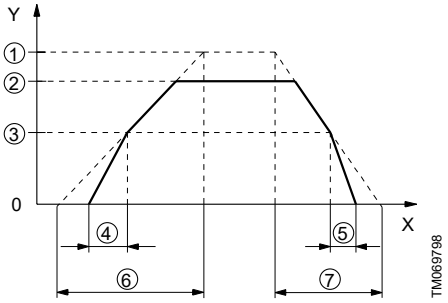
You can make the following settings:

- acceleration time, 0.1 to 300 s
- deceleration time, 0.1 to 300 s.

The times apply to the acceleration from 0 rpm to a fixed maximum speed, and the deceleration from a fixed maximum speed to 0 rpm, respectively.

At short deceleration times, the deceleration of the product may depend on load and inertia as there is no possibility of actively braking the product.

If the power supply is switched off, the deceleration of the product only depends on the load and inertia.



Pos.	Description
Y	Speed
X	Time
1	Fixed maximum
2	User-set maximum
3	User-set minimum
4	Fixed initial ramp
5	Fixed final ramp
6	Ramp time up
7	Ramp time down

8.20 Skip band

Use this function to select a skip band within the range from user-set minimum speed to user-set maximum speed if continuous operation is not required. The upper and lower speeds are stated in percentage of rated speed.

The purpose of the skip band is to avoid certain speeds which may cause noise or vibrations. If no skip band is required, select **-**.

8.21 Standstill heating

Use this function to avoid condensation in humid environments.

When you set the function to **Active** and the product is in operating mode **Stop**, a low AC voltage is applied to the motor windings. The voltage is not high enough to make the motor rotate, but ensures that sufficient heat is generated to avoid condensation in the product, including the electronic parts in the drive.

Remember to remove the drain plugs and fit a cover over the product.

8.22 Alarm handling

This setting determines how the pump must react in case of a sensor failure.

The alarm or warning types are the following:

- **Warning**:

  It is a warning, but there is no change in the operating mode.
- **Stop**:

  The pump stops.
- **Min.**:

  The pump reduces the speed to minimum.
- **Max.**:

  The pump increases the speed to maximum.
- **User defined speed**:

  The pump runs at the speed set by the user.

The affected inputs are the following:

- **Analog input 1**
- **Analog input 2**
- **Analog input 3**
- **Grundfos Direct Sensor**
- **Liqtec input**.

8.23 Motor bearing monitoring

Use this function to select whether or not you want to monitor the motor bearings.

You can make the following settings:

- **Active**
- **Not active**.

When the function is set to **Active**, a counter in the controller starts counting the running hours of the bearings. The running hours are calculated based on the motor speed. When a predefined limit is reached, a warning indicates that the bearings must be replaced or relubricated.

If you change the function to **Not active**, the counter continues to count. However, no warning is given when it is time to replace the bearings. If you change the function to **Active** again, the accumulated running hours are used to recalculate the replacement time.

8.24 Service intervals

Motor bearing monitoring must be activated in order for the motor to indicate that the bearings must be replaced or relubricated. See the section on motor bearing monitoring.

For motors of 7.5 kW and below, it is not possible to relubricate the bearings.

8.24.1 Motor bearing service

This display shows when to replace the motor bearings. The controller monitors the operating pattern of the motor and calculates the period between bearing replacements.

Displayable values:

- **in 2 years**
- **in 1 year**
- **in 6 months**
- **in 3 months**
- **in 1 month**
- **in 1 week**
- **Now!**

8.24.2 Bearing replacements

The display shows the number of bearing replacements made during the lifetime of the motor.

8.24.3 Motor bearing maintenance

When the bearing monitoring function is active, a warning is given when the motor bearings must be replaced.

1. Replace the motor bearings.
2. Press **Bearings replaced**.

8.24.4 Bearings relubricated

When the bearing monitoring function is active, a warning is given when the motor bearings must be relubricated.

Bearings can be relubricated 5 times before they must be replaced.

The amount of grease can be found on the bearing nameplate on the motor.

1. When you have relubricated the bearings, press **Bearings relubricated**.

8.25 Communication

Use this function to set the communication of the product, both wired and wireless communication. The product contains built-in fieldbus protocols on the AYB terminals (RS-485).

8.25.1 Pump number

Use this function to allocate a unique number to the pump. This makes it possible to distinguish between pumps in connection with GENIbus communication.

8.25.2 Enable/disable radio comm.

Use this function to set the radio communication to **Enabled** or **Disabled**. Select **Disabled** in areas where radio communication is not allowed.

Bluetooth communication remains active.

8.25.3 Enable/disable Bluetooth comm.

Use this function to set the Bluetooth communication to **Enabled** or **Disabled**. Select **Disabled** in areas where Bluetooth communication is not allowed.

Radio communication remains active.

8.25.4 Initiate Bluetooth connection

Use this function if Grundfos GO is installed on Huawei smartphones with BLE version 5.0 or older. This function is used to establish a Bluetooth connection to Grundfos GO. Open the Grundfos GO app on your device and select **Initiate Bluetooth connection**. Then select **Yes** and follow the instructions on the device.

8.25.5 Setup of AYB terminals

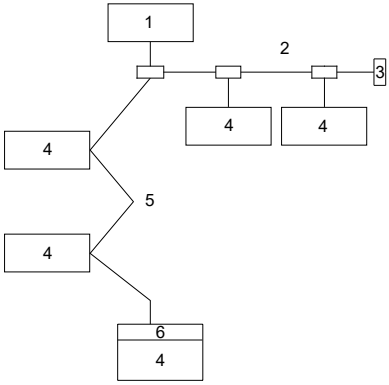
8.25.5.1 Protocol selection

Use this function to select which fieldbus protocol that must be active on the AYB terminals (RS-485).

Select between the following:

- Modbus RTU
- GENIbus.

8.25.5.2 Modbus RTU settings

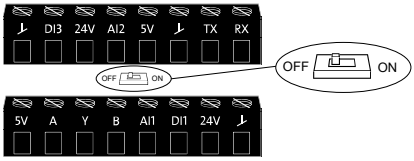


TM083380

Example of Modbus network with termination

Pos.	Description
1	Master
2	Passive tap
3	Line termination
4	Slave
5	Daisy chain
6	BLT (BLT = Built-in line termination (dip switch))

Remember to set the AYB BUS termination dip switch to ON in case the pump is the first or the last pump on a daisy chain of pumps. The termination resistor has a value of 150 ohm.



TM083381

Modbus RTU address

Use this function to allocate a unique number to the pump. This makes it possible to distinguish between pumps in connection with Modbus RTU communication.

Select a number between 1 and 247.

Baud rate

Use this function to select the baud rate at which Modbus RTU is to communicate.

Select between the following baud rates:

- 9600 bps
- 19200 bps
- 38400 bps
- 115200 bps.

Parity

Use this function to set the parity of the Modbus RTU channel.

Select between these values:

- None
- Odd
- Even.

Stop bits

Use this function to set the number of stop bits on the Modbus RTU channel.

Select between these values:

- 1 bit
- 2 bits.

8.25.6 Setup of Ethernet

The product is equipped with an Ethernet port with a GENI GDP protocol that can be accessed from Grundfos iSOLUTION Cloud and other cloud based solutions.

Grundfos will support the product with security updates for at least 2 years from production of the unit.

8.25.6.1 IP Settings

Use this function to set the Ethernet communication.

8.25.6.2 DHCP

Use this function to select if DHCP should be activated or deactivated.

If activated, the E-pump will receive network configuration from the DHCP server on the network.

If deactivated, IP address, Subnet mask, Gateway and Primary DNS must be configured manually.

8.25.6.3 IP address

Use this function to manually set the IP address. IP address format:

Example: 192.168.0.10

8.25.6.4 Subnet mask

Use this function to manually set the subnet mask. Subnet mask format:

Example: 255.255.255.0

8.25.6.5 Gateway

Use this function to manually set the gateway address. Gateway address format:

Example: 192.168.1.1

8.25.6.6 Primary DNS

Use this function to manually set the primary DNS address.

Example of primary DNS address format: 8.8.8.8

8.25.6.7 Secondary DNS

Use this function to manually set the secondary DNS address.

Example of secondary DNS address format: 4.4.4.4

8.26 Language

The function is only available in the HMI 300 and HMI 301 operating panels.

Use this function to select the desired language from the list.

8.27 Set date and time

The function is only available in the HMI 300 and HMI 301 operating panels.

Use this function to set the date and time as well as how you want them to be viewed on the display.

- **Select date format**:
  - **YYYY-MM-DD**
  - **DD-MM-YYYY**
  - **MM-DD-YYYY**.
- **Select time format**:
  - **HH:MM 24-hour clock**
  - **HH:MM am/pm 12-hour clock**.
- **Set date**.
- **Set time**.

8.28 Units

This function is only available in the HMI 300 and HMI 301 operating panels.

Use this function to select SI or US units. You can make the setting for all parameters or customize for each individual parameter.

8.29 Enable/disable settings

Use this function to disable the adjusting of settings for protective reasons.

- If you use Grundfos GO and set the buttons to **Not active**, the buttons on the HMI 200 or HMI 201 operating panel are disabled, except the **Radio communication** button.
- If you disable the buttons on pumps fitted with the HMI 300 or HMI 301 operating panel via **Enable/ disable settings**, you can still use the buttons to navigate through the menus but you cannot make changes directly on these operating panels. A lock symbol appears on the display. However, you can unlock the motor temporarily and allow settings by pressing the **Up** and **Down** buttons simultaneously for at least 5 seconds.

8.30 Delete history

This function is only available in the HMI 300 and HMI 301 operating panels.

Use this function to delete the following historical data:

- **Delete operating log**
- **Delete energy consumption**.

8.31 Define Home display

This function is only available in the HMI 300 and HMI 301 operating panels.

Set the **Home** display to show up to four user-defined parameters.

8.32 Display settings

This function is only available in the HMI 300 or HMI 301 operating panels.

Use this function to adjust the display brightness. You can also set whether or not the display is to switch off if no buttons are activated for a period of time.

8.33 Store actual settings

Use this function to store the current settings to enable the user to go back to a previous set of settings.

8.34 Recall stored settings

Grundfos GO

In this menu, you can recall stored settings from a number of previously stored settings that the pump then uses.

Advanced operating panel

In this menu, you can recall the last stored settings that the pump then uses.

8.34.1 Undo

This function is only available in the Grundfos GO.

Use this function to undo all settings made with the Grundfos GO in the current communication session. Once you recall settings, you cannot undo it.

8.35 Pump name

This function is only available in the Grundfos GO.

Use this function to give the motor a name. The selected name then appears in the Grundfos GO.

8.36 Connection code

Use the connection code to enable automatic connection between the Grundfos GO and the product. Thus, you do not need to press **OK** or the **Radio communication** button each time.

You can also use the connection code to restrict remote access to the product.

You can only set the connection code with the Grundfos GO.

8.36.1 Setting a connection code in the product by using the Grundfos GO

1. Connect Grundfos GO to the product.
2. Go to **Settings** > **General** > **Connection code**.
3. Enter a connection code and press **OK**.
   You can change the code in the **Connection code** menu at any time. The old code is not required.

8.37 Run start-up guide

The function is only available in the HMI 300 and HMI 301 operating panels.

The startup guide automatically starts when you start the product for the first time. You can always run it later. It guides you through the general settings of the product.

To run the startup guide, go to **Settings** > **General settings** > **Run start-up guide**.

8.38 Alarm log

This function contains a list of logged alarms from the product. The log shows the alarm code, name of the alarm, when the alarm occurred and when the alarm was reset.

8.39 Warning log

This function contains a list of logged warnings from the product. The log shows the warning code, name of the warning, when the warning occurred and when the warning was reset.

8.40 Assist

This menu consists of a number of different assist functions.

Assist functions are small guides that take you through the steps needed to set the product.

8.41 Assisted pump setup

This function guides you through the following:

Setting the motor

- Selection of control mode
- Configuration of feedback sensors
- Adjustment of the setpoint
- Controller settings
- Summary of settings.

With Grundfos GO, access the **Assisted pump setup** menu.

With the HMI 300 or 301 operating panel, access the **Assisted pump setup** menu.

Related information

8.5.1 Constant pressure

8.7 Grundfos Direct Sensor

8.42 Setup, analog inputs

This function is only available in the HMI 300 and HMI 301 operating panels.

- **Analog inputs**, follow on-screen instructions.

8.43 Setting of date and time

The function is only available in the HMI 300 and 301 operating panels.

The inputs and outputs available depend on the functional module fitted in the motor.

Functional module	Setting of date and time
FM300	●
FM310	●
FM311	●

The function guides you through the following settings:

- **Select date format**
- **Set date**
- **Select time format**
- **Set time**.

8.44 Description of control mode

This function is only available in the HMI 300 and HMI 301 operating panels.

The function describes each of the control modes available for the product.

8.45 Assisted fault advice

This function provides guidance and corrective actions in the event of product failure.

8.46 Priority of settings

With the Grundfos GO, you can set the motor to operate at maximum speed or to stop.

If two or more functions are enabled at the same time, the motor operates according to the function with the highest priority.

If you set the motor to maximum speed via the digital input, the motor operating panel or the Grundfos GO can only set the motor to **Manual** or **Stop**.

The priority of the settings is shown in the table below.

Priority	Start/stop button	Grundfos GO or operating panel	Digital input	Bus communication
1	**Stop**			
2		**Stop** [15)]		
3		**Manual**		
4		**Maximum speed** [15)]**/User defined speed**		
5			**Stop**	
6			**User defined speed**	
7				**Stop**
8				**Maximum speed/User defined speed**
9				**Minimum speed**
10				**Start**
11			**Maximum speed**	
12		**Minimum speed**		
13			**Minimum speed**	
14			**Start**	
15		**Start**		

15) **Stop** and **Maximum speed** settings made with the Grundfos GO or the operating panel can be overruled by another operating-mode command sent from a bus, for example **Start**. If the bus communication is interrupted, the motor resumes its previous operating mode, for example **Stop**, selected with the Grundfos GO or the operating panel.

8.47 Factory settings for Grundfos GO

Settings	With factory-fitted sensor	Without factory-fitted sensor
Setpoint	67 % of sensor range	67 % speed
Operating mode	**Normal**	**Normal**
Set user-defined speed	67 %	67 %
Control mode	**Constant pressure**	**Constant curve**
Pipe-filling function	**Not active**	**Not active**
Buttons on product	**Active**	**Active**
Stop function (**Low-flow stop function**)	**Not active**	**Not active**
Controller	**Kp**: 0.5 **Ti**: 0.5	**Kp**: 0.5 **Ti**: 0.5
Operating range	9-100 %	9-100 %
Ramps	**Ramp-up time**: 0.5 s **Ramp-down time**: 3 s	**Ramp-up time**: 0.5 s **Ramp-down time**: 3 s
Number	1	1
Radio communication	**Enabled**	**Enabled**
Analog input 1	**4-20 mA**	**Not active**
Analog input 2	**Not active**	**Not active**
Analog input 3	**Not active**	**Not active**
Pt100/1000 input 1	**Not active**	**Not active**
Pt100/1000 input 2	**Not active**	**Not active**
Digital input 1	**Ext. stop**	**Ext. stop**
Digital input 2	**Not active**	**Not active**
Digital input/output 3	**Not active**	**Not active**
Digital input/output 4	**Not active**	**Not active**
Pulse flowmeter (**Pulse flowmeter setup**)	-	-
Predefined setpoint	0 bar	0 %
Analog output	**Speed/0-10 V**	**Speed/0-10 V**
External setpoint funct.	**Not active**	**Not active**
Signal relay 1	**Alarm**	**Alarm**
Signal relay 2	**Ready**	**Ready**
Limit 1 exceeded	**Not active**	**Not active**
Limit 2 exceeded	**Not active**	**Not active**
Limit 3 exceeded	**Not active**	**Not active**
Limit 4 exceeded	**Not active**	**Not active**
LiqTec (**LiqTec function**)	**Not active**	**Not active**
Detection delay	**10 seconds**	**10 seconds**
Standstill heating	**Not active**	**Not active**
Motor bearing monitoring	**Not active**	**Not active**

Settings	With factory-fitted sensor	Without factory-fitted sensor
Pump name	-	-
Connection code	-	-
Unit configuration (Units)	SI	SI

Related information

8.7 Grundfos Direct Sensor

8.13 Controller settings

9. Servicing the product

WARNING
Electric shock
Death or serious personal injury

- Switch off the power supply to the product including the power supply for the signal relays. Wait at least 5 minutes before you make any connections in the terminal box. Make sure that the power supply cannot be switched on accidentally.
- Tighten the cable glands to the recommended torques.



- For measuring power supply voltage, use the measuring points accessible through the holes on the cover for power cables.
- Follow the instructions in the service instructions for the motor. If parts are damaged, order new service kits.
- Connect the motor to protective earth and provide protection against indirect contact in accordance with local regulations.
- After servicing the motor, a dielectric strength test must be performed. Alternatively, a megger can be used at 500 VDC.

WARNING
Rotating parts
Death or serious personal injury

- Stay clear of the product after switching on power, as the shaft can rotate immediately.



- Do not start up and run the motor if there is no pump connected to it.
- Install the coupling guards securely to the pump with the screws intended for this purpose.
- Tighten the coupling screws to the correct torque.



WARNING
Magnetic field
Death or serious personal injury

- Do not handle the motor or rotor if you have a pacemaker.



WARNING
Crushing of hands
Death or serious personal injury

- Follow the instructions in the service instructions for the motor.
- Wear protective gloves when servicing the product.
- Be careful when handling magnetised parts to avoid personal injury.



WARNING
Falling objects
Death or serious personal injury

- Follow the lifting instructions for the product.
- Use lifting equipment rated for the weight of the product.



WARNING
Back injury
Death or serious personal injury

- Use lifting equipment and follow local regulations when lifting the product.



WARNING
Crushing of feet
Death or serious personal injury

- Wear safety shoes.
- When lifting the motor, attach lifting equipment to the eye bolts fitted to the motor. When lifting the terminal box, attach lifting equipment to the eye bolts or lifting brackets fitted to the terminal box.



WARNING
Hot surface
Death or serious personal injury

- Do not touch the product while it is running. Allow surfaces to cool before servicing.



WARNING
Intoxication or risk of chemical burn
Death or serious personal injury

- The battery can cause severe or fatal injuries in 2 hours or less if it is swallowed or placed inside any part of the body. In such an event, seek medical attention immediately.



- The replacement or servicing of batteries must be carried out by a qualified person.
- The battery contained within this product, whether new or used, is hazardous and is to be kept away from children.



CAUTION
Cold surface
Minor or moderate personal injury

- Make sure that no one can accidentally come into contact with cold surfaces. Wear protective gloves.



Do not remove the rotor from the motor.



Make sure to fill the pump with water before the power is switched on. Follow the instructions for the pump.

Related information

3.3 Lifting the product
12.5.7 Torques for terminals

9.1 Insulation resistance test



Do not use a megger or similar high-voltage instrument, as the built-in electronics may be damaged.

9.2 Maintenance

9.2.1 Cleaning the product



WARNING
Electric shock
Death or serious personal injury

- Switch off the power supply to the product including the power supply for the signal relays. Make sure that the power supply cannot be switched on accidentally.
- Check that the terminal box cover is intact before spraying water or chemicals on the product.
- Cleaning must be done with non-aggressive materials to avoid damage to surfaces and labels.
- Make sure that the air inlets are kept clean and free of residuals.



Do not expose the product to high-pressure water jets.

To clean the motor, follow the procedure below:

1. Let the motor cool down first to avoid condensation.
2. Spray it with cold water, and use only non-aggressive cleaning materials.

10. Taking the product out of operation



WARNING
Electric shock
Death or serious personal injury

- Switch off the power supply and make sure that it cannot be accidentally switched on. The power supply must be switched off for at least five minutes before you start working on the product.



WARNING
Back injury
Death or serious personal injury

- Use lifting equipment and follow local regulations when lifting the product.



The lifting eyes on the motor can be used for lifting the pump as well.



For lifting instructions, see the related installation and operating instructions for the pump.

11. Fault finding



WARNING
Electric shock
Death or serious personal injury

- Switch off the power supply before you start any work on the product.
- Make sure that the power supply cannot be switched on accidentally.



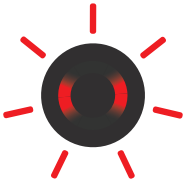
For information on fault finding, see the related installation and operating instructions for the pump.

Related information

5.2.9 Signal relays
7.4 Grundfos Eye

11.1 Fault and warning signals

The fault indication below can be read by the number in brackets, text or status indications on the operating panel, depending on the equipment available.

Grundfos Eye	Cause	Remedy
 TM053839 Two opposite red indicator lights flashing simultaneously. (Alarm indication)	**External fault (3)**	
	An external signal reports an external fault to the digital input set for this function.	• Check the parameter or the unit reporting the external fault. • Correct the fault.
	Too many restarts (4)	
	The pump restarts too many times due to a fault that forces the pump to stop and restart automatically.	• Check the warning and alarm log for faults that caused too many restarts. • Replace the pump if the fault cannot be found.
	Overvoltage (32)	
	The supply voltage to the pump is too high.	• Make sure that the power supply is within the specified range.
	Undervoltage (40)	
	The supply voltage to the pump is too low.	• Make sure that the power supply is within the specified range.
	Undervoltage transient from mains supply (41)	
	The supply voltage to the pump shows signs of voltage drops.	• Make sure that the power supply is within the specified range and check the power quality.
	Overload (49)	
	The motor is overloaded and has automatically reduced the speed, thereby reducing the pump performance.	• Check if the viscosity and temperature of the pump liquid is within the limits for the pump. If not, change the properties of the liquid. • Disassemble the pump, and remove any foreign objects or impurities preventing the pump from rotating. • If none of the above causes are present, replace the pump.
	Blocked pump (51)	
	The pump is blocked.	• Disassemble the pump, and remove any foreign objects or impurities preventing the pump from rotating.
	STO Active Indication (62)	
	The Safe Torque Off (STO) function are activated by an external device.	• This might not be an alarm situation. The pump might be stopped on purpose due to the activation of STO by an external device. • Verify STO functionality according to the STO manual.

Grundfos Eye	Cause	Remedy
 TM053835 One yellow indicator light permanently on. (Warning indication)	**Pump communication fault (10)**	
	There is a communication fault between this pump and the other pumps of the multi-pump system.	• Make sure that all pumps of the multi-pump system are correctly set.
	Forced pumping (29)	
	Other pumps or sources force flow through the pump even if the pump is stopped.	• Check the system for defective non-return valves and replace, if necessary. • Check the system for correct position of, for example, non-return valves.
	Dry running (56, 57)	
	There is no water at the pump inlet, or the water contains too much air.	• Prime and vent the pump before a new startup. • Check if the pump operates correctly. If not, replace the pump.
	Internal fault (72, 83, 85, 155, 157, 163)	
	There is an internal fault in the pump electronics.	• Replace the functional module, power board or terminal box.
	High motor temperature (65, 66)	
	The motor temperature is too high.	• Make sure that the ambient temperature is within the specified range. • Make sure that the pump is not covered by dust, dirt or other foreign objects which reduce the air cooling of the pump. • If none of the above causes are present, replace the motor.
	Internal communication fault (76)	
	There is a communication fault between different parts of the electronics.	• Replace the terminal box.
	Soft pressure buildup, timeout (215)	
	The system is in the mode "soft pressure buildup" longer than the set time limit.	• Check the system for leakages.

Grundfos Eye	Cause	Remedy
	Replace the motor bearings (30)	
	The bearings are worn.	• Replace the bearings. Follow the instructions for the pump. See the relevant section on bearing replacement in the service instructions.
	Internal sensor fault (88)	
	The pump receives a signal from the internal sensor which is out of the normal range.	• Check that the plug and cable are connected correctly to the sensor. The sensor is on the back of the pump housing. • Replace the sensor.
	Pt100/1000 sensor 1 (91) and 2 (175)	
	Pt100/1000 input 1 receives a signal which is out of the normal range.	• Check if the sensor resistance is approximately 100 or 1000 ohm. If not, replace the sensor. • Check the sensor cable for damage. • Check the cable connection of the pump and the sensor. Correct the connection, if required. • Replace the sensor.
 TM053833 One yellow indicator light rotating in the direction of rotation of the motor when seen from the non-drive end. (Warning indication)	**Supply fault, 5 V (161)**	
	Fault in the output voltage to the sensor or potentiometer.	• Check the output voltage and wire to the sensor or potentiometer.
	Supply fault, 24 V (162)	
	Fault in the output voltage.	• Check the output voltage and wire.
	Signal fault, LiqTec-sensor (164)	
	The pump receives a signal from the LiqTec sensor which is out of the normal range.	• Check if the plug and cable are connected correctly to the sensor. • Replace the sensor.
	Signal fault, sensor 1 (165), 2 (166) and 3 (167)	
	Analog input 1, 2 or 3 receives a signal which is out of the normal range.	• Check if the setup of the analog input corresponds to the sensor output as regards electrical characteristics (0.5 - 3.5 V, 0-5 V, 0-10 V, 0-20 mA or 4-20 mA). If not, change the setting, or replace the sensor with one that matches the setup. • Check the sensor cable for damage. • Check the cable connection of the pump and the sensor. Correct the connection, if required. • Check if the sensor is removed, but the input is not deactivated. • Replace the sensor.
	Limit 1 exceeded (190) and limit 2 exceeded (191)	

Grundfos Eye	Cause	Remedy
	Limit 1 or 2 reaches the limit for warning or alarm.	• Identify and remove the fault cause.

12. Technical data

12.1 Operating conditions

12.1.1 Installation altitude

The installation altitude is the height above sea level of the installation site.

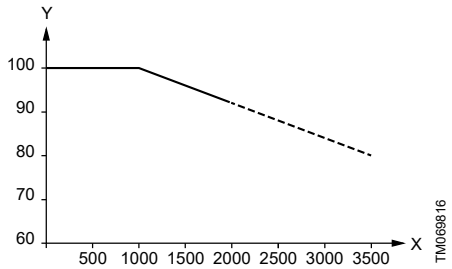
Products installed up to 1000 m above sea level can be loaded 100 %.

The motors can be installed up to 3500 m above sea level.

Model I

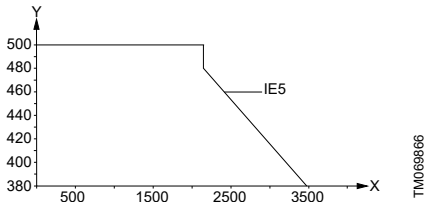
Products installed more than 1000 m above sea level must not be fully loaded due to the low density and consequent low cooling effect of the air.

The motor output power (P2) in relation to the altitude above sea level is shown in the graph.



Pos.	Description
Y	Motor output power [%]
X	Altitude [m]

To maintain the galvanic isolation and ensure correct clearance according to EN60664-1:2007, adapt the supply voltage to the altitude. The supply voltage for a three-phase motor in relation to the altitude is shown in the graph.



Pos.	Description
Y	Supply voltage
X	Altitude [m]

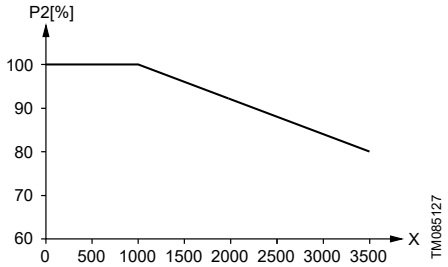
Model J and model K

Model J: The product is not suitable for use on corner earthed grids in installations more than 2000 m above sea level.

Model K: The product is not suitable for use on corner earthed grids. The motors can be installed up to 3500 m above sea level.

Products installed more than 1000 m above sea level must not be fully loaded due to the low density and consequent low cooling effect of the air.

The motor output power (P2) in relation to the altitude above sea level is shown in the graph.



Pos.	Description
P2	Motor output power [%]
X	Altitude [m]

12.1.2 Maximum number of starts and stops

The number of starts and stops via the power supply must not exceed ten times per hour.

When switched on via the power supply, the product starts after approximately 5 seconds.

If a higher number of start and stops are required, use a digital input for external start and stop when starting and stopping the product or use the Safe Torque Off (STO) function.

When started via an external on and off switch, the product starts immediately.

12.1.3 Ambient temperature

12.1.3.1 Ambient temperature during storage and transportation

Description	Temperature
Minimum	-30 °C
Maximum	60 °C

12.1.3.2 Ambient temperature during operation

Description	3 × 200-240 V	3 × 380-500 V [16]
Minimum	-20 °C	-20 °C
Maximum	40 °C	40 °C

16) The motor can operate with the rated power output (P2) at 50 °C. Continuous operation at higher temperatures reduces the expected product life. If the motor operates at ambient temperatures between 50 and 60 °C, select an oversized motor. Contact Grundfos for further information.

12.1.4 Humidity

Description	Percentage
Maximum humidity (non-condensing)	95 %

If the humidity is constantly high and above 85 %, open the drain holes in the drive-end flange to vent the motor.

If you install the motor in moist surroundings or areas with high humidity, ensure that the bottom drain hole is open. As a result, the motor becomes self-venting, allowing water and humid air to escape. When you open the drain hole, the enclosure class of the motor will be lower than standard.

12.1.5 Turbine operation

Do not force the product to run at a higher speed than the maximum speed stated on the nameplate.

12.2 Technical data, three-phase motors

DANGER
Electric shock
Death or serious personal injury
- Use the recommended fuse size.

Supply voltage

- 3 × 200-240 V - 10 %/+ 10 %, 50/60 Hz, PE
- 3 × 380-480 V -10 % / +10 %, 50/60 Hz, PE
- 3 × 380-500 V - 10 %/+ 10 %, 50/60 Hz, PE
- 3 × 400-480 V -10 % / +10 %, 50/60 Hz, PE.

Check that the supply voltage and frequency correspond to the values stated on the nameplate.

Recommended size of fuse or circuit breaker

You can use standard as well as quick-blow or slow-blow fuses.

Model I

Supply voltage	Motor size [kW]	Minimum [A]	Maximum [A]
3 × 200-240 V	1.5	10	20
3 × 380-500 V	1.5	6	10
	2.2	6	16

Model J

Supply voltage	Motor size [kW]	Minimum [A]	Maximum [A]
3 × 200-240 V	2.2	13	35
	3	16	35
	4	25	35
	5.5	32	35
3 × 380-500 V	3	10	16
	4	13	16
	5.5	16	32
	7.5	20	32
	11	32	32

Model K

Supply voltage	Motor size [kW]	Recommended [A]	Maximum[A]	Fuse type
3 × 200-240 V	7.5	60	80	gG
	11	70	80	gG
3 × 380-480 V	15	50	80	gG
	18.5	60	80	gG
	22	70	80	gG
3 × 400-480 V	26	80	80	gG

12.2.1 Leakage current (AC)

The leakage currents are measured without any load on the shaft and in accordance with EN 61800-5-1:2007.

- 3 × 380-500 V - 10 %/+ 10 %, 50/60 Hz, PE
- 3 × 200-240 V - 10 %/+ 10 %, 50/60 Hz, PE.

The leakage currents are measured without any load on the shaft and in accordance with EN 61800-5-1:2007.



WARNING
Electric shock
Death or serious personal injury

- If the leakage current is greater than 3.5 mA, use a PE cable with a minimum cross-section of at least 10 mm^2, or use 2 separate PE cables with the same cross section as the power cable.

Model I, 50/60 Hz

Supply voltage [V]	Power [kW]	Mains voltage[V]	Leakage current[mA]
3 × 380-500 V	1.5 - 2.2	≤ 400	< 3.5
		> 400	< 5

Model J, 50/60 Hz

Supply voltage [V]	Power [kW]	Mains voltage[V]	Leakage current (I_L) [mA]
3 × 200-240 V	2.2 - 5.5	200-240	< 3.5
3 × 380-500 V	3 - 5.5	≤ 400	< 3.5
		> 400	< 3.5
	7.5 - 11	≤ 400	< 3.5
		> 400	3.5 < I_L < 5.0

Model K, 50/60 Hz

Supply voltage [V]	Power [kW]	Mains voltage[V]	Leakage current (I_L) [mA]
3 × 200-240 V	7.5 - 11	> 200	3.5 < I_L < 20
3 × 380-480 V	15-22	≤ 400	3.5 < I_L < 20
		> 400	3.5 < I_L < 30
3 × 400-480 V	26	≤ 400	3.5 < I_L < 20
		> 400	3.5 < I_L < 30

12.3 Inputs and outputs

Signal reference

All voltages refer to signal ground (GND). All currents return to signal ground.

Absolute maximum voltage and current limits

Exceeding the following electrical limits may result in severely reduced operating reliability and motor life.

Relay 1:

- Maximum contact load: 250 VAC, 2 A or 30 VDC, 2 A.

Relay 2:

- Maximum contact load: 30 VDC, 2 A.

GENI terminals: -5.5 to +9.0 VDC or less than 25 mADC.

Other input and output terminals: -0.5 to +26 VDC or less than 15 mADC.

Digital inputs

Internal pull-up current greater than 10 mA at V_i equal to 0 VDC.

Internal pull-up to 5 VDC. Currentless for V_i greater than 5 VDC.

Input activated level: V_i less than 1.5 VDC.

Input deactivated level: V_i from 3.0 VDC to 24 VDC.

Hysteresis: No.

Screened cable: 0.5 - 1.5 mm^2 / 28-16 AWG.

Maximum cable length: 500 m.

Safe Torque Off (STO) terminals

S24:

24 V output voltage. Only for use with ST1 and ST2 inputs.

- Output voltage: 24 V -5 % to +5 %
- Maximum current: 50 mADC
- Overload protection: Yes.

ST1 and ST2:

- STO activated: V_{in} lower than 1.25 V
- STO deactivated: V_{in} greater than 21.6 V and lower than 25 V
- Input current greater than 10 mA at V_{in} equal to 24 V.

When the internal voltage source (connection S24) is used, the input voltage for ST1 and ST2 is within accepted limits.

When an external voltage source is used to drive the STO inputs, the following conditions must be met:

In operational state, the input voltage of ST1 and ST2 with reference to GND must be within:

- V_{min}: 21.6 V
- V_{max}: 25.0 V.

In the safe state, the input voltage of ST1 and ST2 with reference to GND must be as follows:

- V_{max}: 1.25 V.

In the operating state, the current flow into ST1 and ST2 must be within:

- Minimum contact current: 10 mA
- Maximum contact current: 25 mA.

Input source rating: SELV

Bus input (Ethernet)

Protocols TC/IP GENI, GDP.

Cable type, Standard CAT5, CAT5e or CAT6.

Open-collector digital outputs (OC)

Current-sinking capability: 75 mADC, no current sourcing.

Load types: Resistive and/or inductive.

Low-state output voltage at 75 mADC: Maximum 1.2 VDC.

Low-state output voltage at 10 mADC: Maximum 0.6 VDC.

Overcurrent protection: Yes.

Screened cable: 0.5 - 1.5 mm^2 / 28-16 AWG.

Maximum cable length: 500 m.

Analog inputs (AI)

Voltage signal ranges:

- 0.5 - 3.5 VDC, AL AU
- 0-5 VDC, AU
- 0-10 VDC, AU.

Voltage signal:

- R_i greater than 100 kΩ at 25 °C.

Leak currents may occur at high operating temperatures. Keep the source impedance low.

Current signal ranges:

- 0-20 mADC, AU
- 4-20 mADC, AL AU.

Current signal: R_i is equal to 292 Ω.

Current overload protection: Yes. Change to voltage signal.

Measurement tolerance: +/- 2 % of full scale.

Screened cable: 0.5 - 1.5 mm^2 / 28-16 AWG.

Maximum cable length: 500 m, excluding potentiometer.

Potentiometer connected to +5 V, GND, any AI: Use maximum 10 kΩ.

Maximum cable length: 100 m.

Analog output (AO)

Current sourcing capability only.

Voltage signal:

- Range: 0-10 VDC
- Minimum load between AO and GND: 1 kΩ
- Short-circuit protection: Yes.

Current signal:

- Ranges: 0-20 and 4-20 mADC
- Maximum load between AO and GND: 500 Ω
- Open-circuit protection: Yes.

Tolerance: +/- 4 % of full scale.

Screened cable: 0.5 - 1.5 mm^2 / 28-16 AWG.

Maximum cable length: 500 m.

Grundfos Digital Sensor input and output (GDS)

Use Grundfos Digital Sensor only.

Power supplies, +5 V, +24 V

+5 V

- Output voltage: 5 VDC -5 % to +5 %
- Maximum current: 60 mADC, sourcing only
- Overload protection: Yes.

+24 V

- Output voltage: 24 VDC -5 % to +5 %
- Maximum current: 200 mADC, sourcing only
- Overload protection: Yes.

Digital outputs, relays

Potential-free changeover contacts.

Minimum contact load when in use: 5 VDC, 10 mA.

Screened cable: 0.5 - 2.5 mm^2 / 28-12 AWG.

Maximum cable length: 500 m.

Bus input

Grundfos GENIbus protocol, RS-485.

Grundfos Modbus protocol, RS-485.

Screened 3-core cable: 0.5 - 1.5 mm^2 / 28-16 AWG.

Maximum cable length: 500 m.

12.4 Accessories

Communication interface modules intended for use with the product are listed in the following table.

Protocol	Communication interface module
GENIbus	CIM 050
LonWorks (Single)	CIM 100
LonWorks (Multi)	CIM 110
PROFIBUS DP	CIM 150
Modbus RTU	CIM 200
BACnet MS/TP	CIM 300
Modbus TCP, BACnet IP, PROFINET, GiC/GRM IP, EtherNet IP	CIM 500
Ethernet GiC	CIM 550
Modbus TCP, Straton T5	CIM 550 P

Installing a communication interface module not listed above might affect the compliance level of the product.

12.5 Other technical data

12.5.1 Ecodesign Directive

This product is out of scope of Directive 2009/125/EC and Commission Regulation (EU) 2019/1781 due to Article 2 (3a), as the variable speed drive (VSD) is integrated into a product and its energy performance cannot be tested independently from the product.

Related information

5.2.6.2 Cable glands

12.5.2 EMC (electromagnetic compatibility)

Standard used: EN 61800-3.

The table below shows the emission category of the motor.

C1 fulfils the requirements for residential areas.

Note: When connected to a public network, 11 kW motors do not comply with the partial weighted harmonics (PWH) requirements of EN 61000-3-12. If required by the distribution network operator, compliance can be obtained in the following way:

The impedance of the mains cables between the motor and the point of common coupling (PCC) must be equivalent to the impedance of a 50 m cable with a cross-section of 0.5 mm.

C3 fulfils the requirements for industrial areas.

Note: When the motors are installed in residential areas, supplementary measures may be required to prevent the motors from causing radio interference.

Motor [kW]	Emission category
	2900-4000 min^{-1}
1.5	C1
2.2	C1
3	C1
4	C1
5.5	C1
7.5	C3/C1 [17)]
11	C3/C1 [17)]

17) C1, if equipped with an external Grundfos EMC filter.

Immunity: The motor fulfils the requirements for industrial areas.

Contact Grundfos for further information.

12.5.3 Insulation class

311 °F (155 °C).

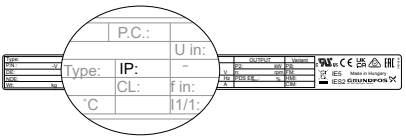
12.5.4 Enclosure class

Standard: IP55.

Optional: IP66.

IP ratings are evaluated in accordance with IEC 60034-5.

The IP rating can be found on the product nameplate:



TM084099

12.5.5 Standby power consumption

5-10 W.

12.5.6 Cable gland sizes

Model I

Motor [kW]	Quantity	Thread size	Cable diameter [mm]
1.5 - 2.2	2	M20 × 1.5	5
	1		7-14

Model J

Motor [kW]	Quantity	Thread size	Cable diameter [mm]
2.2	2	M20 × 1.5	3-9
	1		7-14
3 - 5.5	4	M20 × 1.5	3-9
	1	M25 × 1.5	9-18
7.5 - 11	5	M20 × 1.5	3-9
	1	M32 × 1.5	14-25

Model K

Motor [kW]	Quantity	Thread size	Cable diameter [mm]
7.5 - 26	4	M20 × 1.5	3-9
	1	M40 × 1.5	16-28

Related information

5.2.6.2 Cable glands

12.5.7 Torques for terminals

Model I

Terminal	Thread size	Maximum torque [Nm]
L1, L2, L3, L, N	M4	1.8
NC, C1, C2, NO	M2.5	0.5
1-26, A, Y, B	M2	0.5

Model J

Terminal	Recommended torque [Nm]
L1, L2, L3	2.2
PE	6
NC, C1, C2, NO	0.5
DI1, DI2, DI3, DI4, AI1, AI2, AI3, AO1, PT1, PT2, LT1, LT2, GND, 24V, 5V, TX, RX, A, Y, B, S24, ST1, ST2	0.5

Torques for other parts

Part designation	Recommended torque [Nm]
Control box, upper part	6.5 - 7
Cover for mains	1.0 - 1.3
Cable glands:	
M20/M40	1 - 1.5
Blind plugs:	
M20	1 - 1.5
½" NPT	8 - 10

Related information

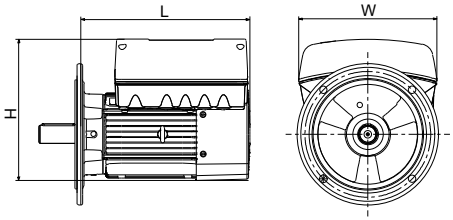
5.2.6 Cable requirements
9. Servicing the product

12.5.8 Sound pressure level

Motor [kW]	Rated maximum speed [min^{-1}]	Speed [min^{-1}]	Sound pressure level ISO 3743 [dB(A)]
			3-phase motor
1.5	4000	3000	57
		4000	64
2.2	4000	3000	57
		4000	64
3	4000	3000	60
		4000	69
4	4000	3000	61
		4000	69
5.5	4000	3000	61
		4000	69
7.5	4000	3000	66
		4000	73
11	4000	3000	66
		4000	73

12.5.9 Dimensions and weights

Motor with free-hole flange (FF), B5



TM069878

Model I

Supply voltage [V]	Power [kW]	L [mm]	W [mm]	H [mm]	Weight [kg]
3 × 200-240	1.5	274	268	223	15.0
3 × 380-500	1.5	274	268	223	15.0
	2.2	274	268	223	16.4

Model J

Supply voltage [V]	Power [kW]	L [mm]	W [mm]	H [mm]	Weight [kg]
3 × 200-240	2.2	334	291	298	22
	3.0	334	291	298	22
	4.0	334	291	298	25
	5.5	389	346	367	37
3 × 380-500	3.0	334	291	298	24
	4.0	334	291	298	26
	5.5	365	291	298	35
	7.5	389	346	367	41
	11	406	346	367	58

Model K

Supply voltage [V]	Power [kW]	L [mm]	W [mm]	H [mm]	Weight [kg]
3 × 200-240	7.5	482	420	462	87
	11	482	420	462	87
3 × 380-480	15	482	420	462	87
	18.5	482	420	462	91
	22	508	420	462	106
3 × 400-480	26	508	420	462	106

13. Disposing of the product

This product or parts of it must be disposed of in an environmentally sound way.

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.
3. Dispose of the waste battery through the national collective schemes. If in doubt, contact your local Grundfos company.



The crossed-out wheelie bin symbol on a product means that it must be disposed of separately from household waste. When a product marked with this symbol reaches its end of life, take it to a collection point designated by the local waste disposal authorities. The separate collection and recycling of such products will help protect the environment and human health.

See also end-of-life information at *www.grundfos.com/product-recycling*.

14. Document quality feedback

To provide feedback about this document, scan the QR-code using your phone's camera or a QR code app.

Click here to submit your feedback

A.1. Installation in the USA and Canada

To maintain the cURus approval, the additional information in this section must be followed.

The UL approval is according to UL 1004-1.

Environmental enclosure ratings

For more information about ambient temperature during operation, see the sections on operating conditions and ambient temperature.

EMC statements for USA

This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

MLE motors of the C1 emission category fulfill the limits of Class B.

MLE motors of the C3 emission category can only be used in industrial plants and public utilities in accordance with FCC § 15.103(b) and ICES 003 § 1.5.1(c). In other locations, MLE motors of the C1 or C2 emission category must be used.

Canadian Interference-Causing Equipment Standard

This product complies with the Canadian ICES-003 Class B specifications. This Class B device meets all the requirements of the Canadian interference-causing equipment regulations.

Cet appareil numérique de la Classe B est conforme à la norme NMB-003 du Canada. Cet appareil numérique de la Classe B respecte toutes les exigences du règlement sur le matériel brouilleur du Canada.

A.1.1. Electrical codes

For the USA

This product complies with the Canadian Electrical Code and the US National Electrical Code.

This product has been tested according to the national standards for Electronically Protected Motors:

CSA 22.2 100-14:2014 (applies to Canada only).

UL 1004-1:2015 (applies to USA only).

Pour le Canada

Codes de l'électricité:

Ce produit est conforme au code canadien de l'électricité et au code national de l'électricité américain.

Ce produit a été testé selon les normes nationales s'appliquant aux moteurs protégés électroniquement:

CSA 22.2 100.04: 2009 (s'applique au Canada uniquement).

UL 1004-1: Juin 2011 (s'applique aux États-Unis uniquement).

A.1.2. Radio communication

For the USA and Canada

CAUTION
Radiation
Minor or moderate personal injury

- This equipment complies with FCC and ISED radiation exposure limits set forth for an uncontrolled environment. This equipment must be installed and operated with a minimum distance of 8 inches (20 cm) between the radiator and your body.

This device complies with Part 15 of the FCC rules and RSS210 of the IC rules.

Changes or modifications made to this equipment not expressly approved by Grundfos may void the FCC authorization to operate this equipment.

Operation is subject to the following two conditions:

- This device may not cause interference.
- This device must accept any interference, including interference that may cause undesired operation of the device.

Pour les États-Unis et le Canada

CAUTION
Radiation
Minor or moderate personal injury

- Cet équipement est conforme aux limites d'exposition aux rayonnements définies par la FCC et l'ISDE pour un environnement non contrôlé. Cet équipement doit être installé et utilisé à une distance minimale de 20 cm (0,66 pi) entre le radiateur et votre corps.

Cet appareil est conforme à la section 15 de la réglementation FCC et à RSS210 de la réglementation IC.

Les changements ou modifications apportés à cet équipement qui ne sont pas expressément approuvés par Grundfos peuvent annuler l'autorisation de la FCC à utiliser cet équipement.

Son fonctionnement est soumis aux deux conditions suivantes:

- Ce dispositif ne doit pas provoquer de brouillage préjudiciable.
- Il doit accepter tout brouillage reçu, y compris le brouillage pouvant entraîner un mauvais fonctionnement.

A.1.3. Identification numbers

For the USA

Grundfos Holding A/S
Contains FCC ID: OG3-RADIOM01-2G4
Contains FCC ID: OG3-RA2G4MSR.

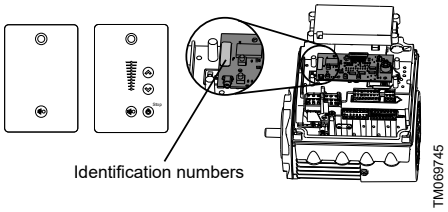
For Canada

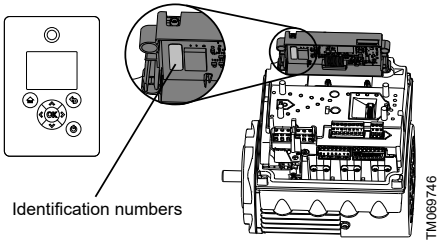
Grundfos Holding A/S
Model: RADIOMODULE 2G4
Contains IC: 10447A-RA2G4M01
Contains IC: 10447A-RA2G4MSR.

Pour le Canada

Numéros d'identification:
Grundfos Holding A/S
Modèle: RADIOMODULE 2G4
Contient IC: 10447A-RA2G4M01
Contient IC: 10447A-RA2G4MSR.

Location of identification numbers





A.1.4. Electrical connection

Installation altitude

For 480/277V grid systems: The maximum altitude is between 0 and 3500 m above sea level.

Torques

See the section on torques.

Conductors

See the sections on electrical installation and cable requirements.

Conductor temperature ratings

Use minimum 60 °C copper conductors.

Ethernet cable connection

The connection of Ethernet cables must be done by connecting the Ethernet cable screen to an earth clamp on the terminal box, to be in compliance with FCC and ISED requirements.

The recommended Ethernet cable types for earth clamp applications are SF/UTP, S/FTP or SF/FTP, where the cable screen consists of both a braided and a foil screen.

Torques

See the section on torques.

Line reactors

The maximum line reactor size in front of the drive must not exceed the following values:

Model I

P2		Maximum line reactor size
[kW]	[hp]	[mH]
1.5 - 2.2	2 - 3	1.5

Model J

P2		Maximum line reactor size
[kW]	[hp]	[mH]
2.2	3	1.5
4	5	0.7
5.5	7.5	0.3
7.5	10	0.6
11	15	0.3

Model K

P2		Maximum line reactor size
[kW]	[hp]	[mH]
11	15	-
15	20	0.2
18.5	25	0.2
22	30	0.2
26	35	0.2

Exceeding these values creates resonance between the reactor and the drive, which reduces the life of the product.

Fuses

Fuses used for motor protection must be rated for minimum 500 V. Motors up to and including 10 HP require class K5 UL-listed fuses. Any UL-listed fuse can be used for motors of 15 HP.

Short-circuit current

If a short circuit occurs, the motor can be used on a mains supply delivering not more than 5000 RMS symmetrical amperes, 600 V maximum.

Branch-circuit protection

When the pump is protected by a circuit breaker, the circuit breaker must be rated for a minimum voltage of 500 V. The circuit breaker must be of the inverse-time type.

Overload protection

Degree of overload protection provided internally by the drive, in percent of full-load current: 102 %.

B.1. Instalação no Brasil

B.1.1. Comunicação via rádio

Este equipamento está em conformidade com a Parte 15 das normas da FCC, IC – RSS-210 e da Anatel com número de registro 01507-14-7763. A operação é de acordo com as duas condições:

- Este equipamento não pode causar interferência.
- Este equipamento deve aceitar qualquer interferência, incluindo interferência que pode causar operação indesejada do equipamento.

Os usuários são alertados de que qualquer alteração ou mudança não expressamente aprovada pela Grundfos pode anular o direito da operação do equipamento pelo usuário.

Este equipamento não tem direito à proteção contra interferência prejudicial e não pode causar interferência em sistemas devidamente autorizados.

B.1.2. Números de Identificação

Grundfos Holding A/S

Contém FCC ID: OG3-RADIOM01-2G4.

Contém IC: 10447A-RA2G4M01.

Modelo: RADIOMODULE 2G4.

Contém Anatel: 01507-14-7763.

Appendix C

C.1. Intended use in the United Kingdom

Safety precautions for UK:



The product is not intended for use in any home appliance, home automation, home control system or consumer product in the UK.

China RoHS

1. 中国 RoHS

产品中有害物质的名称及含有信息表

部件名称	有害物质									
	铅 (Pb)	汞 (Hg)	镉 (Cd)	六价铬 (Cr6+)	多溴联苯 (PBBs)	多溴二苯醚 (PBDEs)	邻苯二甲酸二正丁酯（DBP）	邻苯二甲酸二异丁酯 (DIBP)	邻苯二甲酸丁基苄酯（BBP）	邻苯二甲酸二（2-乙基）己酯（DEHP）
印刷电路板	X	O	O	O	O	O	O	O	O	O
紧固件	X	O	O	O	O	O	O	O	O	O

注 1：O：表示该有害物质在该部件所有均质材料中的含量均不超出电器电子产品有害物质限制使用国家标准要求。
X：表示该有害物质至少在该部件的某一均质材料中含量超出电器电子产品有害物质限制使用国家标准要求。
注 2：以上未列出的部件，表明其有害物质含量均不超出电器电子产品有害物质限制使用国家标准要求。

TM090586

Declaration of conformity

GB: EU declaration of conformity

We, Grundfos, declare under our sole responsibility that the products MGE and MLE - models J, to which the declaration below relates, are in conformity with the Council Directives listed below on the approximation of the laws of the EU member states.

CZ: EU prohlášení o shodě

My společnost Grundfos prohlašujeme na svou plnou odpovědnost, že výrobky MGE and MLE - models J, YYY, na které se toto prohlášení vztahuje, jsou v souladu s níže uvedenými ustanoveními směrnice Rady pro sblížení právních předpisů členských států Evropské unie.

DE: EU-Konformitätserklärung

Wir, Grundfos, erklären in alleiniger Verantwortung, dass das Produkt MGE and MLE - models J, auf das sich diese Erklärung bezieht, mit den folgenden Richtlinien des Rates zur Angleichung der Rechtsvorschriften der EG-/EU-Mitgliedsstaaten übereinstimmt.

DK: EU-overensstemmelseserklæring

Vi, Grundfos, erklærer under ansvar at produkterne MGE and MLE - models J som erklæringen nedenfor omhandler, er i overensstemmelse med Rådets direktiver der er nævnt nedenfor, om indbyrdes tilnærmelse til EU-medlemsstaternes lovgivning.

ES: Declaración de conformidad UE

Grundfos declara, bajo su exclusiva responsabilidad, que los productos a los que concierne la presente declaración, marcados con la denominación MGE and MLE - models J, son conformes con las Directivas del Consejo que figuran a continuación, basadas en la aproximación de las legislaciones correspondientes de los Estados miembros de la UE.

FR : Déclaration de conformité UE

Nous, Grundfos, déclarons sous notre entière responsabilité que les produits MGE and MLE - models J auxquels la déclaration ci-dessous fait référence, sont conformes aux directives du Conseil répertoriées ci-dessous, concernant le rapprochement des législations des États membres de l'UE.

HU: EU megfelelőségi nyilatkozat

Mi, a Grundfos vállalat, teljes felelősséggel kijelentjük, hogy a(z) MGE and MLE - models J termékek, amelyre az alábbi nyilatkozat vonatkozik, megfelelnek az Európai Unió tagállamainak jogi irányelveit összehangoló tanács alábbi előírásainak.

IT: Dichiarazione di conformità UE

Grundfos dichiara sotto la sua esclusiva responsabilità che i prodotti MGE and MLE - models J, ai quali si riferisce questa dichiarazione, sono conformi alle seguenti direttive del Consiglio riguardanti il riavvicinamento delle legislazioni degli Stati membri UE.

NL: EC Conformiteitsverklaring

Wij, Grundfos, verklaren geheel onder eigen verantwoordelijkheid dat de producten MGE and MLE - models J, waarop de onderstaande verklaring betrekking heeft, in overeenstemming zijn met de onderstaande Richtlijnen van de Raad inzake de onderlinge aanpassing van de wetgeving van de EU-lidstaten.

PL: Deklaracja zgodności WE

Firma Grundfos oświadcza z pełną odpowiedzialnością, że jej produkty MGE and MLE - models J, których dotyczy niniejsza deklaracja, są zgodne z następującymi dyrektywami Rady w sprawie zbliżenia przepisów prawnych państw członkowskich UE.

SE: EU-försäkran om överensstämmelse

Vi, Grundfos, försäkrar under ansvar att produkterna MGE and MLE - models J, som omfattas av nedanstående försäkran, är i överensstämmelse med de rådsdirektiv om inbördes närmande till EU-medlemsstaternas lagstiftning som listas nedan.

SK: EÚ vyhlásenie o zhode

My, spoločnosť Grundfos, vyhlasujeme na svoju plnú zodpovednosť, že produkty MGE and MLE - models J, na ktoré sa vyhlásenie uvedené nižšie vzťahuje, sú v súlade s ustanoveniami nižšie uvedených smerníc Rady pre zblíženie právnych predpisov členských štátov EÚ.

TR: AB uygunluk beyanı

Grundfos olarak, aşağıdaki bildirim konusu olan MGE and MLE - models J ürünlerinin, AB üye ülkelerinin direktiflerinin yakınlaştırılmasıyla ilgili aşağıdaki Konsey Direktifleriyle uyumlu olduğunu ve bununla ilgili olarak tüm sorumluluğun bize ait olduğunu beyan ederiz.

- All products

  – RoHS Directives (2011/65/EU) and (2015/863/EU)

  Standard used:

  EN IEC 63000:2018
- Products marked: HMI 100, HMI 200 or HMI 300 and FM110 or FM310

  – Radio Equipment Directive (2014/53/EU) and (2022/30/EU)

  Standards used: EN 61800-5-1:2007+A1:2017+A11:2021, EN 61800-3:2018, EN 62311:2008, EN 301 489-1 V2.2.3, EN 301 489-17 V2.3.4, EN 300 328 V2.2.2, EN 18031-1:2024
- Products marked: HMI 101, HMI 201 or HMI 301 and FM311

  – Low Voltage Directive (2014/35/EU)

  Standards used: EN 61800-5-1:2007+A1 :2017+A11 :2021, EN 60034-1 :2010+AC :2010

  – EMC Directive (2014/30/EU)

  Standard used: EN 61800-3:2018

This EU declaration of conformity is only valid when accompanying Grundfos instructions.

Bjerringbro, 13 March, 2025



Jakob Fredsted
Sr. Director of PD Motors and Drives
Grundfos Holding A/S
Poul Due Jensens Vej 7
8850 Bjerringbro, Denmark
www.grundfos.com

Person authorised to compile technical file and empowered to sign the EU declaration of conformity.

Declaration of conformity

GB: EU declaration of conformity

We, Grundfos, declare under our sole responsibility that the products MGE160, MGE180, MLE160, MLE180 – model K, to which the declaration below relates, are in conformity with the Council Directives listed below on the approximation of the laws of the EU member states.

CZ: EU prohlášení o shodě

My společnost Grundfos prohlašujeme na svou plnou odpovědnost, že výrobky MGE160, MGE180, MLE160, MLE180 – model K, YYY, na které se toto prohlášení vztahuje, jsou v souladu s níže uvedenými ustanoveními směrnice Rady pro sblížení právních předpisů členských států Evropské unie.

DE: EU-Konformitätserklärung

Wir, Grundfos, erklären in alleiniger Verantwortung, dass das Produkt MGE160, MGE180, MLE160, MLE180 – model K, auf das sich diese Erklärung bezieht, mit den folgenden Richtlinien des Rates zur Angleichung der Rechtsvorschriften der EG-/EU-Mitgliedsstaaten übereinstimmt.

DK: EU-overensstemmelseserklæring

Vi, Grundfos, erklærer under ansvar at produkterne MGE160, MGE180, MLE160, MLE180 – model K som erklæringen nedenfor omhandler, er i overensstemmelse med Rådets direktiver der er nævnt nedenfor, om indbyrdes tilnærmelse til EU-medlemsstaternes lovgivning.

ES: Declaración de conformidad UE

Grundfos declara, bajo su exclusiva responsabilidad, que los productos a los que concierne la presente declaración, marcados con la denominación MGE160, MGE180, MLE160, MLE180 – model K, son conformes con las Directivas del Consejo que figuran a continuación, basadas en la aproximación de las legislaciones correspondientes de los Estados miembros de la UE.

FR : Déclaration de conformité UE

Nous, Grundfos, déclarons sous notre entière responsabilité que les produits MGE160, MGE180, MLE160, MLE180 – model K auxquels la déclaration ci-dessous fait référence, sont conformes aux directives du Conseil répertoriées ci-dessous, concernant le rapprochement des législations des États membres de l'UE.

HU: EU megfelelőségi nyilatkozat

Mi, a Grundfos vállalat, teljes felelősséggel kijelentjük, hogy a(z) MGE160, MGE180, MLE160, MLE180 – model K termékek, amelyre az alábbi nyilatkozat vonatkozik, megfelelnek az Európai Unió tagállamainak jogi irányelveit összehangoló tanács alábbi előírásainak.

IT: Dichiarazione di conformità UE

Grundfos dichiara sotto la sua esclusiva responsabilità che i prodotti MGE160, MGE180, MLE160, MLE180 – model K, ai quali si riferisce questa dichiarazione, sono conformi alle seguenti direttive del Consiglio riguardanti il riavvicinamento delle legislazioni degli Stati membri UE.

NL: EC Conformiteitsverklaring

Wij, Grundfos, verklaren geheel onder eigen verantwoordelijkheid dat de producten MGE160, MGE180, MLE160, MLE180 – model K, waarop de onderstaande verklaring betrekking heeft, in overeenstemming zijn met de onderstaande Richtlijnen van de Raad inzake de onderlinge aanpassing van de wetgeving van de EU-lidstaten.

PL: Deklaracja zgodności WE

Firma Grundfos oświadcza z pełną odpowiedzialnością, że jej produkty MGE160, MGE180, MLE160, MLE180 – model K, których dotyczy niniejsza deklaracja, są zgodne z następującymi dyrektywami Rady w sprawie zbliżenia przepisów prawnych państw członkowskich UE.

SE: EU-försäkran om överensstämmelse

Vi, Grundfos, försäkrar under ansvar att produkterna MGE160, MGE180, MLE160, MLE180 – model K, som omfattas av nedanstående försäkran, är i överensstämmelse med de rådsdirektiv om inbördes närmande till EU-medlemsstaternas lagstiftning som listas nedan.

SK: EÚ vyhlásenie o zhode

My, spoločnosť Grundfos, vyhlasujeme na svoju plnú zodpovednosť, že produkty MGE160, MGE180, MLE160, MLE180 – model K, na ktoré sa vyhlásenie uvedené nižšie vzťahuje, sú v súlade s ustanoveniami nižšie uvedených smerníc Rady pre zblíženie právnych predpisov členských štátov EÚ.

TR: AB uygunluk beyanı

Grundfos olarak, aşağıdaki bildirim konusu olan MGE160, MGE180, MLE160, MLE180 – model K ürünlerinin, AB üye ülkelerinin direktiflerinin yakınlaştırılmasıyla ilgili aşağıdaki Konsey Direktifleriyle uyumlu olduğunu ve bununla ilgili olarak tüm sorumluluğun bize ait olduğunu beyan ederiz.

- All products

  – RoHS Directives (2011/65/EU) and (2015/863/EU)

  Standard used:

  EN IEC 63000:2018
- Products marked: HMI 100, HMI 200 or HMI 300 and FM110 or FM310

  – Radio Equipment Directive (2014/53/EU) and (2022/30/EU)

  Standards used:

  EN 61800-5-1:2007+A1:2017+A11:2021, EN 61800-3:2018, EN 62479:2010, EN 301 489-1 V2.2.3, EN 301 489-17 V3.2.4, EN 300 328 V2.2.2, EN 18031-1:2024
- Products marked: HMI 101, HMI 201 or HMI 301 and FM311

  – Low Voltage Directive (2014/35/EU)

  Standards used:

  EN 61800-5-1:2007+A1:2017+A11:2021, EN 60034-1:2010+AC:2010

  – EMC Directive (2014/30/EU)

  Standard used:

  EN 61800-3:2018

This EU declaration of conformity is only valid when accompanying Grundfos instructions.

Bjerringbro, 13 March, 2025



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Declaration of conformity

Standard products

- Machinery Directive (2006/42/EC).
  Standards used:
  EN 809:1998 + A1:2009.
- Ecodesign Directive (2009/125/EC).
  Water pumps:
- Commission Regulation No 547/2012.
  Applies only to water pumps marked with the minimum efficiency index MEI. See pump nameplate.

All products

- RoHS Directives (2011/65/EU and 2015/863/EU).
  Standard used:
  EN IEC 63000:2018

Declaration of conformity related to the motor is enclosed in this manual(publication number 98247034, 92898130).

This EC/EU declaration of conformity is only valid when published as part ofthe Grundfos safety instructions (publication number 98247034, 92898130) and Grundfos installation and operating instruction (publication number 99955183, 99688961).

Bjerringbro, 24th July, 2025



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